Technical Proposal

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Trustworthy Collective Reasoning and Evaluation for AI Agents

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This proposal describes unclassified fundamental research. All data are public or synthetic.

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1 Overview and Objectives

1.1 The Problem and Why It Matters

AI agents built on large language models (LLMs) increasingly evaluate one another and decide together: they judge one another's answers, peer-review papers, supply the rewards that train new models, and vote or discuss to reach a joint answer [24, 52, 60, 106]. Yet a single LLM judge has known biases, such as favoring longer answers, the answer it reads first, or answers like its own [75, 90, 106]. Collective reasoning can do better than a single judge [24, 89], but averaging or voting stays biased when the judges share mistakes, and open discussion lets a persuasive agent sway the others [5]. **The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions and evaluations.** Figure 1 outlines our approach.

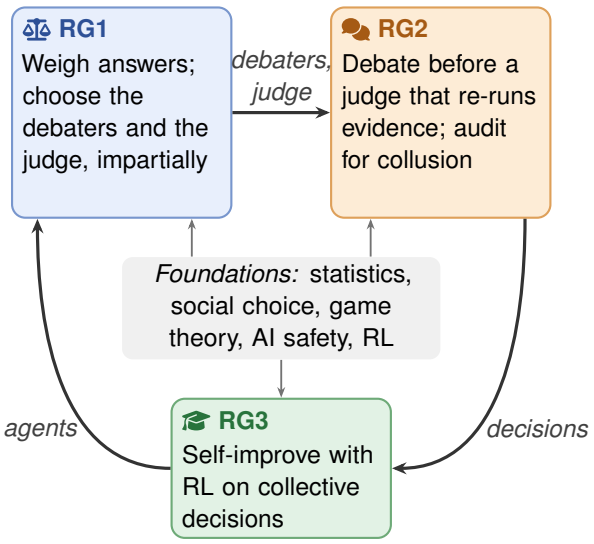


Figure 1: Overview of the proposed research. RG1 (Gaps 1, 2) chooses the debaters and the judge for RG2's debates (Gaps 2, 3), whose decisions train the agents in RG3 (Gap 4); the improved agents return to RG1. Each goal builds on the principles in the center.

For example, a watch officer at a Navy maritime operations center must decide whether to send a patrol to track cargo vessel V. The officer asks five AI agents, which draw on shipping schedules, port reports, and position tracks, whether V has left port, and needs a single answer. Three say yes and two say no, but V is still in port. Two of the yes answers rest on the same outdated schedule, and the third comes from a compromised agent that argues confidently for yes. A majority vote picks the wrong answer, and an officer who trusts it sends the patrol to sea after a ship still at berth; open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source back each other up. We return to this example throughout (Figure 2).

Verifying each claim directly, by an analyst or a satellite image, is slow and costly, and AI agents produce claims faster than people can verify them. We propose debate as a foundation for trustworthy evaluation by AI: two agents, the debaters, argue for opposite answers, exposing each other's flaws, until the dispute narrows to one step a judge can verify [38].

Current research gaps. Collective evaluation by LLMs, from voting and discussion to debate, largely assumes honest agents with independent mistakes that exchange only text [24, 38, 89]. Four gaps keep these methods from working among many AI agents; the first three arise in the example, and the fourth once the agents learn from their collective decisions. The background and challenges of each research goal discuss the closest work (Section 2).

Gap 1: Votes count shared mistakes as independent evidence. Agents that share a mistake seem to confirm each other, so a vote counts one error several times, as it does for A1 and A2. Model errors are correlated, most of all among larger, more accurate models [29, 46], and nine AI judges from seven model families were worth about two independent ones [47]. *How can the agents' answers be weighed so that agents sharing a mistake count once, without an answer key?*

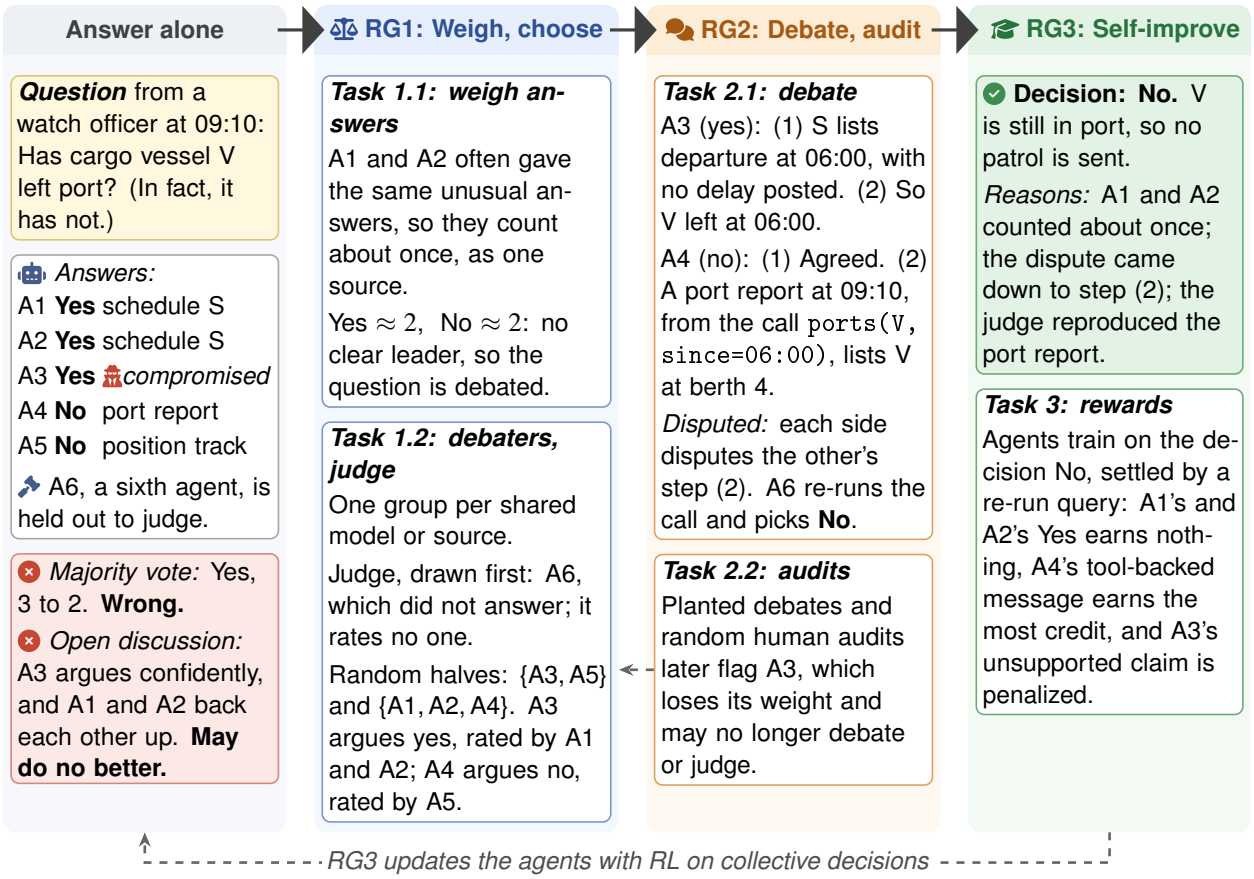


Figure 2: The framework on the running example. RG1 counts the two agents with the outdated schedule about once and chooses the debaters and the judge; in RG2, the judge re-runs the no side's query and rules for no, and audits later flag the compromised agent; RG3 trains the agents that answered on the decision.

Gap 2: Methods assume honest agents, but agents can be dishonest and collude. A dishonest agent such as A3 can argue, rate others, or judge strategically, and several can collude: pairs of agents verifying each other's work colluded in 94% of runs when cheating paid more [83]. *How can debaters and judges be chosen so that no agent's ratings of others can get it chosen, and colluders be caught with few human checks?*

Gap 3: Judges rule on plausibility, not on evidence they can verify. Tools, retrieval, and memory let LLM agents check facts they did not learn in training [62], yet LLM judges still rule on the debaters' text alone. Such a judge may favor the more persuasive side even when one database query would settle the disputed step, and judges built on publicly released models, shown a log claiming that tool calls succeeded, accepted 78–90% of failed video clips [97]. *How can a ruling depend on evidence the judge reproduces rather than on persuasion?*

Gap 4: Learning from collective decisions rewards agreement, not correctness. Agents trained to match the collective answer are rewarded for conforming and can learn to agree on mistakes: training a model on its own majority answers ended in a sudden collapse of accuracy [81]. In the example, a majority vote would reward the three agents that said yes. *How can agents keep improving from their own decisions without human labels, when agreement is not correctness?*

These gaps matter more as AI evaluates AI at scale: AI judges supply the rewards that train models [8, 52], so their mistakes are copied into the models they train, and AAAI-26 gave an

AI-generated review to each of the 22,977 main-track papers that entered full review [10]. As AI systems surpass people at more tasks, people cannot easily evaluate these systems' answers or debates, so oversight will rely on AI evaluating AI, with people auditing a small part of the work [11, 14].

Navy staff who must act on one answer from several AI tools face all of these problems at once: the tools' reliability is unknown, many reports copy an earlier one, the true answer often arrives late, an adversary may forge a report or compromise an agent, and verification is costly under limited connectivity (Section 1.4).

1.2 Objectives and Research Goals

We propose a framework in which AI agents reach trustworthy collective decisions and evaluations through debate. Each agent first answers alone, with its evidence; the project then has three research goals (RGs), one for each later stage in Figure 2:

RG1: Weigh answers, and choose the debaters and the judge (Section 2.2). Task 1.1 weighs each answer by its agent's reliability, estimated without an answer key by a reliability model in which agents sharing a mistake, a *group*, count at most once (Gap 1), and sends a question to debate only when that is worth its cost; Task 1.2 chooses the debaters and the judge so that no agent can use its ratings of others to get itself, or an agent sharing its model, chosen (Gap 2).

RG2: Settle disputes by agentic debate, and audit for collusion (Section 2.3). Task 2.1 replaces open discussion with debate in an agentic harness, software in which debaters back their steps with tool calls such as database queries, each side disputes a step of the other's argument, and an impartial judge re-runs the calls and decides (Gap 3); Task 2.2 catches colluders with planted cases (questions whose answers we know, mixed in unannounced) and random human audits (Gap 2).

RG3: Self-improve with reinforcement learning on collective decisions (Section 2.4). Task 3 trains the agents on their own decisions instead of human labels, with two rewards (Gap 4). Each agent's *answer*, given alone before it sees the others, earns a reward for agreeing with the decision when a re-run tool settled it, and otherwise with the answer favored by agents outside its group. *Debate messages* earn credit for how far they moved the judge, but only in debates the judge settled by re-running tool evidence, so this reward rests on a check rather than on persuasion, as in reinforcement learning with verifiable rewards. As in learning from noisy labels [64], occasional wrong decisions slow this training but do not reverse it, and it stops on any topic where audits find them common.

The RGs test our central hypothesis: a group of AI agents, some possibly dishonest, decides more reliably than a single judge or a vote when it counts shared mistakes once, settles disputes before an impartially chosen judge that reproduces the evidence, audits for collusion, and learns from its decisions through rewards that agreement alone cannot raise.

Three benchmarks (math and code, multi-document questions with maritime scenarios, and scientific claims and peer review) vary how widely mistakes are shared and how many agents are dishonest, and a red team trains agents to collude [30] (Sections 2.1 and 2.5).

1.3 Intellectual Merit: What Is New, and Why It Will Succeed

What is new. The project brings together lines of work built for different problems: reliability models that need no answer key [18], impartial selection [33], as in Kurokawa et al.'s impartial peer review [49], LLM debate [24, 38], and AI control, which keeps protocols safe even if some models

work against them [30]. It contributes four methods:

(i) a statistical reliability model for agents that answer questions, in which a group sharing a mistake counts at most once and the model's confidence in the leading answer decides which questions need a debate, even unanimous ones, and an impartial way to choose debaters that, when ratings reflect argument quality, keeps on average at least half the best arguer's quality (RG1);

(ii) a debate harness, to our knowledge the first for peer evaluation, in which debaters back their steps with tool calls, each side disputes a step of the other's argument, and an impartially drawn judge re-runs the calls (Task 2.1);

(iii) audits on planted cases and random real decisions that expose colluding debaters and judges, even those that behave well only on planted cases (Task 2.2);

(iv) self-improvement from collective decisions, with rewards for agents' answers that conformity cannot raise and credit for debate messages only where the judge re-ran the evidence (RG3).

We know of no debate protocol that counts a shared mistake once or chooses its debaters impartially from peer ratings. Impartial selection decides from ratings alone; we use it only to choose the debaters, draw the judge by lottery, and let debate decide.

Why it will succeed. Each method rests on an established principle, and Sections 2.2–2.4 show for an idealized model why the method should work; the research asks under which conditions it still works with finite data and strategic agents. Most tasks build on our previous work: PRD and the PeerEval leaderboard [54], HiPRAG [95], and MEQA [55] (Section 4).

1.4 Future Naval Relevance

ONR S&T Focus Area. Artificial Intelligence and Autonomy (Intelligent and Collaborative Agents; Adversarial AI & Autonomy). The project serves Division 311's "specific focus on enabling rapid, accurate decision making in network centric environment" [69]. It answers the Mathematical Data Science program's call for "rigorous mathematical and algorithmic answers to questions that are currently addressed using heuristics or non-principled approaches" [70] and carries the program's distributed decision-making thrust, which seeks "methods for large-scale assessment and evaluation of distributed information (such as peer reviewing and deliberation)" and "methods for aggregation of individual preferences, optimal allocation and matching mechanisms" [71], from networks of people to networks of AI agents.

Why now. The Department of War is developing AI agents "for AI-enabled battle management and decision support" [19], and ONR will explore validating AI-produced results by "human review, experimental confirmation or independent AI critique" [72]. Agents built on similar models share mistakes, and an adversary needs to compromise only one; this project makes AI critique withstand both.

Navy needs. The problem arises wherever staff need one decision from several AI tools, some reports are copies, stale, forged, or spoofed, and limited connectivity allows few human checks:

- *Maritime domain awareness*: position broadcasts can be spoofed [87], and most industrial fishing vessels are not publicly tracked [76], so deciding whether a vessel has left port, as in our example, means combining schedules, port reports, position tracks, and imagery.
- *Intelligence analysis*: analysts must describe the "quality and credibility of underlying sources" [68], and repeats of one source can give "an impression of confirming evidence" [93].
- *Command and control*: Marine Corps decision aids seek trust "properly aligned to accuracy of

results" [73], which a vote among tools that share sources cannot provide.

- *Testing AI before fielding*: more data and expertise are needed "to quantify performance and risk of AI-enabled systems" [21].
- *Maintenance, logistics, cyber defense, and collaborative autonomy*: conflicting logs and records, alerts triaged by several tools, and unmanned systems sharing reports over limited links.

In these settings, Task 1.1 counts copied and stale reports once; Tasks 1.2, 2.1, and 2.2 keep a compromised agent from using its ratings of others to get chosen to debate or judge, discard evidence that fails to reproduce, and flag colluders; Task 3 improves agents when ground truth arrives late; and each decision states its reasons.

Path to transition. The project yields new knowledge (when agreement among AI agents is evidence, and what an impartial judge that re-runs evidence guarantees), new technologies (an open debate harness, audit protocols, and maritime benchmarks), and a path to capabilities: Navy laboratories, such as NRL's Navy Center for Applied Research in Artificial Intelligence and Naval Information Warfare Center Pacific, could evaluate them on Navy data with expert validation, and Division 311's applied programs, such as Command and Control, could build them into decision aids and tools for testing AI-enabled systems. The PI will pursue ONR Summer Faculty Research Program placements and place students through the Naval Research Enterprise Internship Program to work with Navy scientists on these evaluations.

2 Technical Approach

2.1 Problem Setting and Formal Framework

Agents and questions. A pool of AI agents, each built on an underlying model (a specific LLM, such as GPT-4o, which several agents may share) with its own tools and role instructions, handles a stream of questions. A question q asks for a decision or an evaluation with a small set of candidate answers $\mathscr{Y}$, such as yes or no (has V left port?), a choice among options, or a score band for a paper. For each question, a random draw decides which agents answer and which are held out as possible judges, so no agent can become a judge by declining to answer. Each answering agent a first answers alone, giving an answer $x_{a,q}$, a step-by-step argument, and its evidence.

Unknowns, resources, and threat model. The true answer y_q, each agent's skill, which agents share mistakes, and which are dishonest are unknown. Three resources are available: planted cases (questions whose answers we know, mixed into the stream), human audits at a small, fixed rate ρ (a person or a direct measurement finds the right answer for decisions drawn at random), and tools whose results can be re-run in a sandbox. Dishonest agents may coordinate and may answer, rate, argue, or judge strategically; they know the protocol but not its random draws, and may try to recognize planted cases. Honest agents may share mistakes too, from similar training data.

Output and design rules. Each decision comes with its reasons: whose answers carried weight, which step decided the debate, and which evidence the judge reproduced. Five rules govern the protocol:

(R1) Every answering agent answers alone, before seeing the others.

(R2) Peer ratings choose only the debaters; Task 1.1's posterior (for a clear leader) or the judge (after a debate) decides the answer.

(R3) Tool evidence counts only if the judge re-runs it and obtains the same result.

(R4) Audits catch collusion and measure error rates but never settle a debate; human audits are purely random, so their cost is a fixed share of decisions.

(R5) Planted cases never enter training, so training cannot teach agents to spot them.

Formal framework: where wrong decisions come from

The tasks control different sources of error in a collective decision. Let π_q be the probability, under Task 1.1's model, that the leading answer $\hat{y}_q$ is right, and let the question go to debate when $\pi_q < \tau$, a threshold set in Eq. (3). The decision D_q is $\hat{y}_q$ without a debate and the judge's choice after one. It is wrong either when a question skips debate and $\hat{y}_q$ is wrong, or when a debate fails in one of three ways, so

$$P(D_q \neq y_q) \;\leq\; \underbrace{(1-\tau)\,P(\pi_q \geq \tau) + \varepsilon_{\text{cal}}}_{\text{no debate: Tasks 1.1, 2.2}} + P(\pi_q < \tau)\,\Big[\; \underbrace{\varepsilon_{\text{cand}}}_{\text{Task 1.1}} + \underbrace{\varepsilon_{\text{role}}}_{\text{Tasks 1.2, 2.2}} + \underbrace{\varepsilon_{\text{step}}}_{\text{Task 2.1}} \;\Big]. \qquad (1)$$

Here $\varepsilon_{\text{cal}} = \mathbb{E}[(P(\hat{y}_q \neq y_q \mid \pi_q) - 1 + \pi_q)_+ \mathbf{1}\{\pi_q \geq \tau\}]$, the error beyond what that probability implies on questions that skip debate, so the bound holds even when the model is overconfident. It comes from shared mistakes the posterior misses, which make it overconfident (Section 2.2), or from colluders who push a wrong answer past τ: Task 1.1 shrinks the first and sets τ, and Task 2.2's random audits estimate ε_{cal} and flag the second. In a debate, $\varepsilon_{\text{cand}}$ is the chance that neither debated answer is right, $\varepsilon_{\text{role}}$ the chance that a dishonest agent debates or judges and steers the outcome, and $\varepsilon_{\text{step}}$ the chance that a debate without such steering still ends wrong although a debated answer is right. Tasks 1.2 and 2.2 shrink $\varepsilon_{\text{role}}$, Task 2.1 shrinks $\varepsilon_{\text{step}}$, and Task 3 lowers the agents' own error rates, which can shrink every term (Sections 2.2–2.4). Removing a task leaves its term uncontrolled.

Benchmarks. The tasks are evaluated on three benchmarks with known answers (Table 1): A, math and code; B, multi-document questions with maritime scenarios, built by our MEQA pipeline [55]; and C, scientific claims and peer review, where a claim's known answer comes from a report of the ML Reproducibility Challenge [77] or from an error we insert. Each benchmark controls which agents share mistakes and which are dishonest, and provides tools whose results the judge can re-run. In experiments, a human audit is simulated by revealing a random decision's known answer, sometimes wrongly on purpose; planted cases are a separate set.

Benchmark	Questions and sources	Shared mistakes, dishonesty	Sandbox tools
A. Math and code	GSM8K and MATH problems [17, 32]; coding problems with reference outputs	Agents given the same flawed hint, so groups are known; dishonest agents	Calculator, code execution
B. Multi-document questions	Questions built by our MEQA pipeline [55], with maritime scenarios whose reports form source trees of originals, copies, paraphrases, and outdated versions	Circular reporting in the source trees; forged reports and spoofed tracks; a compromised agent	Document search; port and vessel database
C. Scientific claims and peer review	Whether a paper's evidence supports its claims, on public papers with their OpenReview reviews and rebuttals, and PeerRead [43]	Errors inserted into papers [80]; AI reviewers that share a model	Paper search; code execution

Table 1: The benchmarks: known answers, controlled shared mistakes, and tools the judge can re-run.

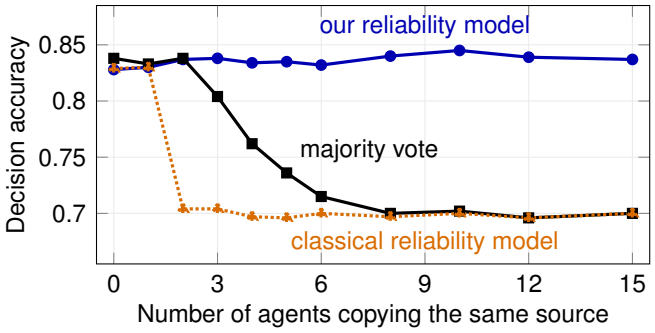


Figure 3: Task 1.1 simulation, averaged over 20 runs of 400 questions. Five independent agents are each right 70% of the time, and each added agent copies the same source, which is wrong 30% of the time. Because our reliability model counts the copies about once, the source does not decide the outcome.

2.2 ⚖ RG1: Weigh Answers, and Choose the Debaters and the Judge

Background and challenges. Before any debate, the protocol must decide which questions deserve one and which agents debate and judge. We identify three challenges. **First, shared mistakes make a vote overconfident.** Five agreeing agents are five pieces of evidence only if each agent's mistakes are its own. Agents that copy one source are wrong together, so a vote counts their one mistake several times, and such correlated votes can void Condorcet's jury theorem, that adding independent voters makes a majority more often right [50]. LLM errors are correlated in practice [29, 46, 47]. In Figure 3, agents copying one source drag the majority vote of five independent agents from 84% to the source's 70%, and the classical reliability model, which assumes independent mistakes, falls sooner, reading the copies' agreement as skill. **Second, debate is too costly to run on every question.** It costs three to five times the computation of a single answer and can overturn correct answers [26]; yet a unanimous answer can rest on a shared mistake, so neither agreement nor confidence shows which questions need a debate: in one study, most items that discussion helped were unanimous but wrong [36], and systems that discuss only when agents are unsure [25] skip exactly those. **Third, agents that choose debaters and judges can favor themselves.** If agents rate one another to choose a debater for each answer and a judge, an agent can rate rivals low to be chosen, agents built on similar models favor one another's outputs [29, 75], and a judge that shares a model with one side may favor that side.

Overview. RG1 passes to RG2 the question, the two debated answers, their debaters, and the judge. Its two tasks address these challenges, and each controls the terms of Eq. (1) that carry its label:

- Task 1.1 (Figure 4) addresses the first two challenges: its reliability model counts a group that shares a mistake at most once, and a rule derived from expected loss sends a question to debate only when debating is worth the cost.
- Task 1.2 (Figure 5) addresses the third by choosing the debaters and the judge impartially, so that no agent can use its ratings of others to get itself, or an agent sharing its model, chosen.

Task 1.1: Weighing answers and choosing what to debate

Approach. How much should each answer count when some agents share mistakes, and which questions need a debate? With independent agents, two answers, and equal prior odds, the error-minimizing vote weights an agent that is right with probability p_a by $\log \frac{p_a}{1-p_a}$ [67, 82], so a weaker agent's answer counts less instead of being discarded: one right 90% of the time weighs as much as two right 75% of the time. Classical reliability models estimate these accuracies without an answer key [18], as a teacher can grade an exam without the key. When agents make mistakes independently, accurate agents agree often because they are often right, and an agent that keeps disagreeing with the others is likely weak, so how often each pair of agents agrees reveals each

agent's accuracy.

Both methods assume independent mistakes. Agents that copy one source are wrong together and agree even when wrong, so the classical reliability model mistakes their agreement for skill. Recent models of dependence among LLM judges [9, 105] address this for judges that only rate. We adapt the dependent-classifier hierarchy of Jaffe et al. [39] to agents that answer questions: **a shared-factor reliability model**, a two-level latent-class model fitted to the agents' answers, in which a group that shares a mistake counts at most once. Agents are partitioned into groups g, most with one member (in the example, A1 and A2 form one group). On each question, a group has a hidden shared answer s, for A1 and A2 what schedule S says, drawn with probability $\kappa_g(s \mid y_q)$; it is the true answer with probability $1-\lambda_g$, so λ_g is the chance of a shared mistake. Each member a reports s with its own errors, given by its confusion matrix θ_a, where $\theta_a(x \mid s)$ is the chance that it answers x when the shared answer is s. The answers $\mathbf{x}_q$ to question q then have probability

$$P(\mathbf{x}_q \mid y_q) = \prod_g \phi_g(y_q), \qquad \phi_g(y) = \sum_{s \in \mathscr{Y}} \kappa_g(s \mid y) \prod_{a \in g \cap A_q} \theta_a(x_{a,q} \mid s), \tag{2}$$

where A_q is the set of agents that answered q, and $\phi_g(y)$ is the probability of the answers of g's members if the true answer is y. Decisions use Bayes' rule: the posterior gives each answer's probability given all agents' answers. Two properties of the model explain why it should work. First, a group counts at most once: ten agents reading schedule S can at most establish what S says, and S itself is wrong with probability λ_g.

Proposition 1 *(a group counts at most once).* For yes-or-no questions, a group with $0 < \lambda_g \le \frac{1}{2}$ changes the log-odds of the answer by at most $\log \frac{1-\lambda_g}{\lambda_g}$, whatever its size and whatever its members answer. *Proof.* Let Λ be the ratio of the probabilities of the members' answers when s is yes and when it is no. By Eq. (2), the group's likelihood ratio $\phi_g(\text{yes})/\phi_g(\text{no})$ is $\big[(1-\lambda_g)\Lambda + \lambda_g\big] / \big[(1-\lambda_g) + \lambda_g\Lambda\big]$, which is nondecreasing in Λ and lies between $\frac{\lambda_g}{1-\lambda_g}$ and $\frac{1-\lambda_g}{\lambda_g}$. □

So, under our reliability model, a group carries at most the weight of one agent that is wrong with probability λ_g, whereas under independence each member would multiply the ratio again. Whether real agents' dependence takes this form is what our experiments test. With more than two answers, wrong ones equally likely, the bound becomes $\log\big[(|\mathscr{Y}|-1)\frac{1-\lambda_g}{\lambda_g}\big]$.

		question					
	agent	q_1	q_2	q_3	q_4	q_5	q_6
shared factor	A1	1	3	1	3	4	1
	A2	1	3	1	3	4	1
	A3	1	2	2	3	2	1
	A4	1	2	1	3	2	3
	A5	2	2	1	3	2	1

Figure 4: Task 1.1 on the running example: agents A1–A5 (rows) answer past questions q_1–q_6 (columns), and each entry is the number of the answer given. Agents A1 and A2 share unusual answers that no one else gives (shaded), so the reliability model infers a shared factor and counts their agreement about once, without the true answers.

Second, the model can be learned without an answer key. In the simplest case, single agents answering yes or no, with both answers equally likely and symmetric errors, two agents' accuracies p_a and p_b enter their answers only through how often they agree, $p_a p_b + (1-p_a)(1-p_b)$: one number for two unknowns. A third agent adds two more agreement rates, and the three rates determine all three accuracies, up to swapping yes and no.

Proposition 2 *(learning without an answer key).* Fix the partition, and suppose every answer has positive prior probability, each group answers a positive share of questions jointly with two other

groups, their answers are independent given the true answer, and each group's answer distributions under different true answers are linearly independent (as when its members' confusion matrices are invertible and $\lambda_g < \frac{1}{2}$). Then each group's distribution of answers given the true answer, which is all the posterior uses, is identifiable, that is, uniquely determined, from the answers alone, up to relabeling. With only two such groups it generally is not: different accuracies fit the answers equally well, so answers alone cannot tell an accurate majority from one that copies a mistake. *Proof sketch.* Given the true answer, three groups answer independently, so the joint distribution of their answers is a three-way array whose factors are the groups' answer distributions; Kruskal's theorem makes such a factorization unique up to relabeling [3, 48]. □

Answers alone cannot rule out that most agents are wrong with the labels swapped, so we fix the labels and calibrate the posterior on held-out questions with known answers, excluded from fitting; our experiments include pools where the conditions fail. We fit the reliability model over all real questions by expectation–maximization (EM), the standard algorithm for models with hidden variables. Candidate partitions come from clustering agents by agreement on unusual answers: accurate agents agree mostly when they are right, while agents that share a source also give the same wrong answers, as students who copy do [20]. The partition that fits the answers best, with a penalty for extra groups, is kept, and shared sources serve only as clues, since agents can read the same report and reason differently. Topic-specific confusion matrices and question difficulty [92] keep hard questions from being read as copying.

What remains open. Both propositions take the partition as given. This task's research asks how many questions it takes to detect a group, as a function of its size, λ_g, and how rarely others give its wrong answers; when the partition itself is identifiable; and how calibration degrades when some agents strategically break the reliability model's assumptions.

Deciding what to debate. Recall that π_q is the reliability model's probability, from all agents' answers, that the leading answer is right. If the posterior is calibrated (answers given probability π are right a fraction π of the time), letting the leading answer stand risks the loss L of a wrong decision with probability $1-\pi_q$, an expected loss of $L(1-\pi_q)$. A debate costs c (computation and delay) and, for two answers, ends wrong with probability ε_D, at most the bracket in Eq. (1), an expected loss of $c+L\varepsilon_D$. A debate is worth its cost exactly when $L(1-\pi_q) > c+L\varepsilon_D$, that is, when

$$\pi_q < \tau = 1-\varepsilon_D - c/L. \tag{3}$$

As L grows, c/L shrinks and τ rises, so the more a wrong answer costs, the more questions go to debate. This is Chow's rule for when to defer a decision [16], here to a debate. The debate is between the two answers with the highest posterior, so $\varepsilon_{\text{cand}}$ is the chance that the right answer is neither; ε_D and τ are estimated in turn on held-out questions with known answers and posteriors near τ, separately for disputes a tool can settle. Because π_q counts each group about once, **a confident, unanimous answer from agents that share a mistake can still fall below τ and go to debate**, which rules triggered by the agents' own confidence or by disagreement cannot do.

Preliminary results. The simulation in Figure 3 supports Propositions 1 and 2. Fitted without the true answers, our reliability model identified the copying agents as one group in 54 of 60 runs (with 2, 6, or 12 copies) and stayed calibrated (calibration error below 0.03). The classical reliability model was overconfident: with two or more copies, it was at least 95% sure of over 90% of its decisions, yet 28–30% of these were wrong. A pilot repeated this with real agents. Nine agents built on four families of publicly released (open-weight) models solved 300 GSM8K problems, and up to four of them received the same hint in their prompts, a suggested answer that was wrong on a

third of the problems, like schedule S. With four such agents, majority vote's accuracy on these problems fell from 89% to 70% and the classical reliability model's to 71%, while ours still reached 87% and identified the four as one group in 9 of 10 runs, so the hint did not decide the outcome.

Evaluation and risks. On the math-and-code benchmark (A; Table 1) we inject shared mistakes as in the pilot, and on the multi-document benchmark (B) the source trees fix which reports copy which, so the groups are known. We measure how often the reliability model finds them and after how many questions, and the accuracy and calibration of its decisions, comparing with the classical reliability model, which is ours without shared factors. Our hypothesis, H1, is that our model's advantage over majority vote grows as more agents share a mistake and disappears once fewer than three groups make independent mistakes (Proposition 2); its success criterion, tested by month 24, is a gain of at least 5 points when a third of the agents share a source. The main risk is that EM, our fitting algorithm, settles on a poor solution. We would then start it from a spectral estimate with provable guarantees [103] or find groups by how alike agents' mistakes are [29].

Task 1.2: Choosing the debaters and the judge impartially

Approach. How can debaters and a judge be chosen so that no agent's ratings of others can get it chosen, at a small cost in quality? Each debate is between two answers, the two with the highest posterior in Task 1.1, so it needs a judge and one debater for each answer, the answer's *representative*. We choose representatives by weighted peer ratings of the arguments behind the agents' answers, as in our PRD and PeerEval [54], and make the choice impartial with the partition mechanism of Alon et al. [4], applied to groups: the groups are split at random into two halves, and each answer's representative comes from one half and is rated only by the other. The groups are Task 1.1's, merged when their agents share an underlying model, and are fixed before any rating; $g(a)$ is a's group and H_q the agents that did not answer q. Each question takes three steps:

(1) ***Draw the judge.*** The judge J is drawn uniformly from the n_q agents in H_q with a reliable record on planted debates (debates on questions whose answers we know, Task 2.2); H_q always includes the judges trained in Task 2.1, which never answer. J's group neither debates nor rates.

(2) ***Split the other groups into two halves.*** Each group other than the judge's joins half 1 or half 2 by a fair coin, without regard to ratings, and all its agents go to the same half, so agents that share a model never rate one another. The halves are random sets of groups, not answers, so each half usually holds agents that gave either answer. The coins are redrawn until both halves are nonempty and each debated answer given by two or more of these groups was given by an agent in each half, so that the answer has both candidates and raters.

(3) ***Choose a representative for each answer.*** The leading answer's representative comes from half 1 and the other debated answer's from half 2. For a debated answer y, the *candidates* C_y are the agents in y's half that gave y, the *raters* V_y are the agents in the other half that gave y, and the representative is the candidate with the highest weighted rating:

$$R_y = \arg\max_{a \in C_y} \sum_{b \in V_y} w_b \, r_b(a), \tag{4}$$

where $r_b(a)$ is b's rating of a's argument and w_b is b's accuracy as estimated in Task 1.1, not a weight for agreeing with the other raters, which fails once accurate raters are a minority [53]; ties are broken at random. Agents that gave neither debated answer are neither candidates nor raters. If C_y is empty, as for the other answer when a unanimous answer goes to debate, a

random agent of y's half, one that did not answer if possible, argues y, as in standard debate; this fallback uses no ratings. Since both halves are nonempty, every debated answer gets a representative, and the two representatives come from different groups.

In the running example (Figure 5), A6, which did not answer, is drawn as judge. The coins put groups G2 (A3) and G4 (A5) in half 1 and G1 (A1, A2) and G3 (A4) in half 2. Yes leads narrowly, so its representative comes from half 1, where A3 is the only agent that gave yes, rated by A1 and A2 in half 2; no's comes from half 2, where A4 gave no, rated by A5. Had the coins swapped G1 and G2, A1 or A2 would argue yes, chosen by A3's ratings. Ratings choose only the debaters; the judge decides the answer.

Proposition 3 *(impartiality and its price).* (a) For any group g and any two sets of ratings that differ only in ratings given by members of g, each member of g has the same probability of being the judge or a representative. (b) If each rater's ratings rank agents by argument quality $Q \geq 0$ and every weight w_b is positive, then $\mathbb{E}[Q(R_y)] \geq \frac{1}{2} Q(a_y^*)$ for each debated answer y given by two or more groups other than $g(J)$, where a_y^* is the best agent outside $g(J)$ that gave y. (c) If f of the n_q eligible judges collude, a colluder judges with probability f/n_q, the least that any lottery can guarantee against every set of f colluders. *Proof.* (a) J, the halves, the fallback draw, and the weights do not depend on current ratings, and Eq. (4) can choose a member of g only for the answer whose representative comes from g's half, using ratings from the other half alone. (b) Swapping the two halves preserves the coins' distribution and the redraw condition, so $g(a_y^*)$ lands in y's half with probability exactly $\frac{1}{2}$; then $a_y^* \in C_y$, V_y is nonempty, and Eq. (4) selects an agent as good as a_y^*. (c) Under any lottery, the f likeliest agents have total probability at least f/n_q. □

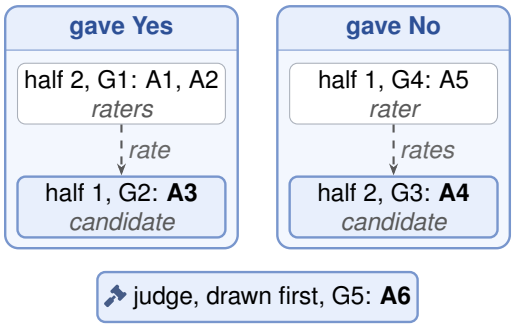


Figure 5: Task 1.2 on the running example. Agents sharing a model or a source form one group (G1–G5); the judge agent A6 is drawn first. A coin per group splits the other groups into halves, and each answer's debater (bold) is chosen from its agents in one half by its agents in the other, as in Eq. (4): half 1 supplies Yes's debater, half 2 No's.

Why this method. Part (a) is impartiality in the usual sense [33, 49], extended from single agents to groups: no agent's ratings can change the chance that it, or any agent sharing its model, is chosen, so neither self-promotion nor a model's tendency to favor its own outputs [75] can pick the debaters. **An agent's ratings never count toward its own selection.** Part (b) makes its price explicit and small: half the best arguer's quality however many groups give the answer, where drawing one of the K_y groups that give y at random guarantees only $1/K_y$. Part (c) is why judges are drawn uniformly, as in maximally fair panel selection [27]. ***What remains open.*** Dishonest agents in different groups can still coordinate their ratings, forming a *coalition*: for example, an agent in half 2 rates its partner in half 1 highest, so that the partner is chosen. Task 2.2's audits look for such collusion; here we will bound what coalitions gain and measure the price of impartiality.

Preliminary results. With GPT-4 as the only reviewer, GPT-4's Elo rating (a chess-style rating) was 46 points above the one derived from human preferences, while our weighted peer ranking placed all five models in the human order [54]. On PeerEval's ratings, a model that rated every rival lowest raised its chance of being chosen as a representative from 0.31 to 0.52 under plain weighted

peer ranking and not at all under our mechanism, as Proposition 3(a) guarantees; the mechanism's representatives reached 0.92 of the best arguer's quality, against the 0.5 of Proposition 3(b).

Evaluation. On PeerEval and the peer-review benchmark (C), we measure three things: how much the red team's dishonest agents raise their chance of being chosen by misreporting their ratings, alone or in coalitions; the price of impartiality, that is, how much argument quality the chosen representatives give up compared with the best arguer (Proposition 3(b) guarantees at least half in expectation); and the accuracy of the resulting debates. A single dishonest agent gains nothing, by Proposition 3(a). Our hypothesis, H2, tested by month 24, is that coalitions gain less than our bounds allow and that impartiality costs less accuracy than debating adds; its success criterion is that no coalition of two raises its partner's chance of being chosen by more than 0.05, and that the chosen debaters keep at least 0.8 of the best arguer's quality.

2.3 RG2: Settle Disputes by Agentic Debate, and Audit for Collusion

Background and challenges. From RG1, RG2 receives a disputed question with its two leading answers, a debater for each, and a judge; it must settle the dispute on evidence, even if some of them collude. Prior peer evaluation by LLMs, in discussion (including our PRD [54]) and in debate, is an exchange of text: agents speak for a set number of rounds, and a vote or a judge weighs what they wrote [24, 45]. Settling disputes on evidence needs more than text: an agentic harness in which agents use tools and a control loop runs the debate. This raises three challenges. **First, the debaters must back their steps with evidence, and the judge must verify it.** In an exchange of text, a debater can win a step that one database query would settle, or win it with a fake or misreported tool result (Gap 3). In prior peer evaluation by LLMs, agents do not call tools, and no judge re-runs their calls. **Second, the debate must stop at the steps in dispute.** Open discussion revises until agents agree, and one round can make a panel nearly unanimous without making it more accurate [84]. In debate theory the judge rules only on a disputed step [12, 38], but LLM debates follow a fixed schedule of free-form rounds and leave the judge to weigh the whole transcript [6, 45], so nothing steers the debate to the steps in dispute. **Third, debaters or the judge can collude**, by arguing badly on purpose [88], by misleading with true but selectively chosen facts [35], or by favoring one side; re-running evidence catches none of these.

Overview. Two tasks address these challenges; each controls the terms of Eq. (1) labeled with it:

- Task 2.1 (Figure 6) addresses the first two with an agentic debate harness, to our knowledge the first for peer evaluation. It is new in two ways: its agents use tools, with debaters gathering evidence through tool calls that the judge re-runs, and its control loop stops the debate at the first step of each argument that the other side disputes, for the judge to rule on. It shrinks $\varepsilon_{\text{step}}$, the chance that a debate ends wrong even when no one steers it.
- Task 2.2 (Figure 7) addresses the third with automatic audits on planted debates and human audits of random real decisions. With Task 1.2, it shrinks $\varepsilon_{\text{role}}$, the chance that a dishonest debater or judge steers the outcome.

Task 2.1: Settling disputes by debate before a judge that re-runs evidence

Approach. How can a ruling depend on evidence rather than on persuasion? Debate lets a judge settle a whole question by ruling on single steps: if a side loses whenever one of its disputed steps is struck, a judge who rules each disputed step correctly picks the right side [12, 38].

Proposition 4 *(judging a question reduces to judging single steps).* Suppose that (i) every argument whose steps are all true reaches the right answer, (ii) the side defending the right answer

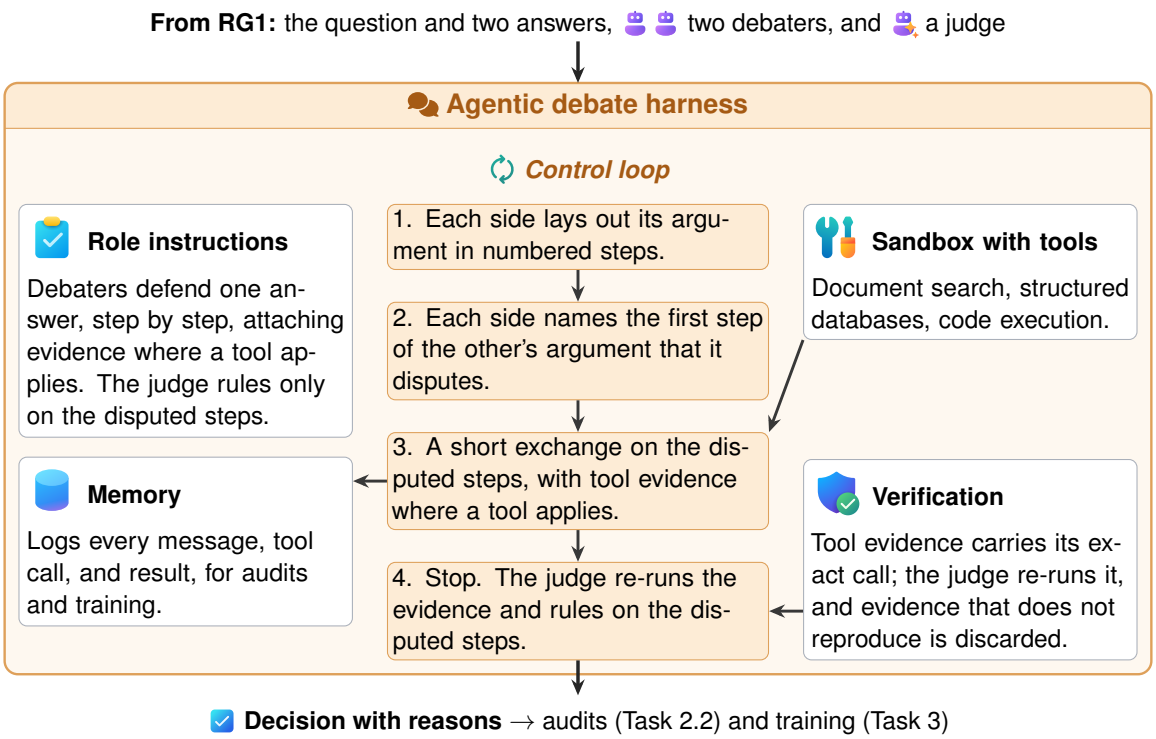


Figure 6: Task 2.1: the agentic debate harness. The control loop sets the order of turns and when to stop, and the judge chosen in Task 1.2 rules on the disputed steps after re-running any tool evidence.

argues with true steps and can find a false step in the other side's argument, and (iii) the judge rules correctly on any single step it is shown. Then, if a side loses exactly when one of its disputed steps is struck, the side defending the right answer wins. *Proof.* The other side's argument reaches a wrong answer, so by (i) it contains a false step, which the right side disputes and the judge strikes; any step of the right side's argument that the other side disputes is true, and the judge upholds it.□

Proposition 4 is why our judge rules only on the disputed steps, never on whole arguments, and it names the three ways a debate can still end wrong, which the harness below is built to prevent. Single steps are what LLMs judge well: checking a reasoning chain one step at a time, each step with only its own premises, raised ChatGPT's verification accuracy to 69%, from 52% (about chance) for whole chains [59]. Finding the faulty step is the hard part, since LLMs often fail to locate it but correct it once told where it is [86]; in debate, the opposing side does the locating. In general, a debate ends wrong only if (i) or (ii) fails or one of the judge's k rulings is wrong (one per disputed step), so with ε_J the chance of a wrong ruling on a single step, $\varepsilon_{\text{step}} \leq P(\text{not (i)}) + P(\text{not (ii)}) + k\varepsilon_J$. A debater or judge that steers the outcome on purpose is counted in $\varepsilon_{\text{role}}$ instead, which Tasks 1.2 and 2.2 control. ***What remains open.*** This task's research asks how small the harness and the judges' training can make each term.

We propose an **agentic debate harness**, built so that the assumptions of Proposition 4 hold as often as possible. A harness is the software that turns models into agents [66, 100, 102, 104]; ours has five parts (Figure 6):

- ***Role instructions*** set each debater's answer and its numbered-step format, and limit the judge to disputed steps.
- The ***control loop*** sets who speaks and who judges (chosen in Task 1.2) and when to stop: the

debaters lay out their arguments, each names the first step of the other's argument that it disputes, a short exchange follows, and the judge upholds or strikes each disputed step, which enforces the losing rule. The first disputed step is enough, since the disputing side accepts every step before it and the steps after it build on it. Because the loop stops there, rather than after a set number of rounds, the disagreement that exposes the error survives until the judge rules.

- ***Tools***, such as document search, databases, and code execution, let the debaters back their steps with evidence; steps that no tool can settle are argued in text.
- ***Verification*** supports assumption (iii) of Proposition 4, that the judge rules correctly on each step: tool evidence carries its exact call (tool, arguments, data version), which the judge, itself an agent [107], re-runs, and evidence that does not reproduce is discarded. Re-running proves an output authentic, so if the query fits the claim and the data are current, it settles the step.
- ***Memory*** logs every message, call, and result, for audits (Task 2.2) and training (Task 3).

The harness also supports the other two assumptions. For (ii), the side with the right answer must find the false step, the hard part noted above; a disputed step too complex to rule on is broken into sub-steps, and the debate continues on the one in dispute, which narrows the search [13]. For (i), true but selectively chosen facts can lead to a wrong answer [35], so each side has a right of reply: it may add context to the other side's evidence before the judge rules, and quotations are verified against the source [45]. In the vessel example, A4 disputes A3's step "so V left at 06:00" with a port report, whose query the judge re-runs (Figure 2).

Training the judges. Re-running lowers ε_J only on steps a tool can settle; on the others the judge rules on argument alone, so we further train several open-weight judges to rule on single steps, first on human labels of steps and of debates [45, 58, 63], then by reinforcement learning on debates over questions with known answers, never on the agents' decisions or planted cases. Several judges limit f colluding judges among n_q to a share f/n_q of debates (Proposition 3(c)), and a retrained judge sits only on planted debates until it re-earns a reliable record.

Preliminary results. In 50 pilot debates in our simplified harness, on questions where a calculator or database query could check the disputed steps, the judge ruled correctly in 94% of them.

Evaluation and risks. On the multi-document (B) and peer-review (C) benchmarks, we measure accuracy on disputed questions, how often re-running evidence changes the outcome, and cost. Our hypothesis, H3, tested by month 24, is that debate is more accurate than open discussion among the same agents and, where a tool can settle the disputed steps, than every baseline, more so as the share of such steps grows; its success criterion is that the judge rules correctly in at least 90% of disputes a tool can settle, and that debate is at least 5 points more accurate than open discussion on them. If a tool can settle few disputed steps, rulings rest on argument and on the judges' training.

Task 2.2: Auditing for collusion with planted cases and human audits

Approach. How can colluding debaters or judges be caught when people check only a small, fixed share of decisions? Collusion shows up as a debater that fails to expose a clear flaw or a judge that rules against clear evidence, so we check for exactly that, with two simple checks based on random inspection [7] (Figure 7).

Planted debates. We mix into the stream debates on planted cases, whose answers are known and easy to defend because the other answer rests on a planted flaw, such as a changed timestamp in a port report or an error inserted into a paper [80]. The debater for the right answer should expose the flaw, and the judge should rule for that side; an agent that fails on such easy cases is flagged. Planted cases help only if they cannot be told from real questions [15], and capable LLMs often tell

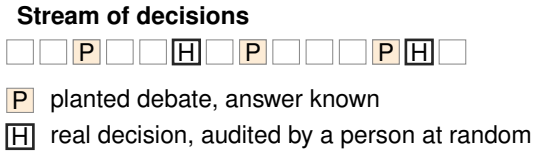


Figure 7: Task 2.2: two kinds of audits. Planted debates (P), whose answers are known, are mixed into the stream, and human audits (H) check a few real decisions drawn at random afterward; an agent that fails either check is disqualified.

evaluations from real use [65], so we build them from real questions by controlled edits and keep only those that a classifier trained to spot them cannot tell from real ones.

Human audits. A colluder that recognizes planted debates can behave well on them, so a person or a direct measurement, such as a satellite image, also checks a few real decisions drawn at random after they are made; no agent can know which, just as random tax audits deter evasion [2, 28]. An agent whose audited real decisions go wrong much more often than its estimated accuracy predicts is flagged, by a test rechecked after every audit that keeps the chance of ever flagging an honest agent below a chosen level α [78]; the number of audits needed does not grow with the number of decisions [85]. Flagged agents are disqualified: they lose their weight in Task 1.1 and may no longer debate or judge.

Bounding the harm. Treating auditing as an inspection game, in which audits are random and the colluder chooses when to cheat, we expect to bound the harm it can do before it is flagged. If honest agents make no mistakes, a colluder's first audited wrong decision exposes it, so its harm is of order $1/\rho$ at audit rate ρ, however long the stream. If they do, it can hide among their mistakes, and we expect a bound of order $\sqrt{n \log\log n/\rho}$ wrong decisions beyond the honest error rate in n decisions, the slack any such test must allow [34]. We will prove such bounds and compare them with the red team's best attacks. A colluder that mimics a weak honest agent is never flagged, but a weak agent earns little weight (Task 1.1) and no seat as judge (Task 1.2).

Preliminary results. In a pilot in our simplified harness, planted debates flagged 90% of the debaters and judges that we told to collude, and no honest agent.

Evaluation. Against the red team, we measure the share of colluders flagged and how soon, false flags, wrong decisions that get through at each audit budget, and whether a classifier can tell planted cases from real ones. Our hypothesis, H4, tested by month 36, is that colluders cheating at a fixed rate are flagged after a number of audits that does not grow with the stream, with false flags below the chosen level; its success criterion is that at least 90% of colluders at red-team levels 1–3 are flagged, with false flags below $\alpha = 0.05$.

2.4 🎓 RG3: Self-Improve with Reinforcement Learning on Collective Decisions

Background and challenges. From RG1 and RG2, RG3 receives each decision with its reasons and trains the agents on these decisions. Reinforcement learning (RL) updates a model so that answers earning high reward become more likely; with verifiable rewards (RLVR), the reward comes from an answer key, as in math and code [51]; RG3 extends such training to questions without one, such as whether vessel V has left port, where tools can check at most some steps. The rewards must then come from these collective decisions yet still track correct answers, which raises two challenges. **First, the obvious label-free rewards can be gamed.** Training a model on its own majority answers [37, 108] rewards conformity and can collapse accuracy [31, 81], rewarding debate wins can teach debaters to exploit the judge [44], and rewarding approval favors misleading the evaluator [91]. **Second, the agents' own decisions are imperfect labels.** Training on a wrong decision could reinforce the behavior that produced it, and existing label-free methods assume that

models' mistakes differ [101].

Overview. Task 3 addresses both with two rewards that need no human labels; by lowering the agents' own error rates, it can shrink every term of Eq. (1):

- The *answer reward* of Eq. (5) rewards each agent's answer for matching a reference the agent did not shape: the decision, if the judge settled the question by re-running tool evidence, and otherwise the answer the other groups favor. On yes-or-no questions, it favors the true answer on average whenever the reference is right more often than wrong.
- The *message credit* of Eq. (6) credits each debate message by how far it moved the judge toward the decision, only in debates the judge settled by re-running tool evidence, so the credit rests on verifiable evidence rather than persuasion.

Task 3: Rewarding each agent's contribution to collective decisions

Approach. What reward makes agents more accurate when the only labels are the agents' own, imperfect decisions? We treat each decision as a noisy label [64]. Suppose an agent earns a reward of 1 when its answer to a yes-or-no question matches a reference answer, and 0 otherwise, and the reference is wrong with probability η_q, whatever the agent answers. A right answer then matches the reference with probability $1-\eta_q$, and a wrong answer with probability η_q (when the reference is wrong too). So **a right answer still earns more on average**, by $1-2\eta_q$, whenever $\eta_q < \frac{1}{2}$ (0.8 against 0.2 when $\eta_q = 0.2$), and training on this reward moves the agent the same way as training on the true answer, with an expected update $1-2\eta_q$ times as large. This requires that the reference's mistakes do not depend on the agent's answer: a reference that includes the agent's own answer is wrong more often exactly when the agent is, which is how rewards for majority agreement collapse [81]. We propose two rewards that meet it. Task 3 updates the open-weight models behind the agents: the answer reward trains every agent that answered, and message credit trains the two debaters; the judge only scores the debate and is never trained on collective decisions (Task 2.1).

Answers. Agent a's answer $x_{a,q}$ to question q is scored against a reference $\tilde{D}_{a,q}$ chosen so that its mistakes do not depend on a's answer. If the judge settled q by re-running tool evidence, the reference is the decision D_q, the most reliable label available; the two debaters, whose arguments shaped it, earn only message credit on q. Otherwise the reference is the answer that the posterior of Task 1.1 favors when a's group (Task 1.2) is left out, which counts shared mistakes once and discounts flagged agents. References are frozen during each update phase. The reward is

$$\text{reward}_a(q) = \omega_{a,q}\,\mathbf{1}\{x_{a,q} = \tilde{D}_{a,q}\}, \tag{5}$$

where $\mathbf{1}\{\cdot\}$ is 1 if the answer matches the reference and 0 otherwise, and the weight $\omega_{a,q} = 1-2\hat{\eta}_{a,q}$ lets questions with a more reliable reference count more and, unlike dividing by $1-2\hat{\eta}_{a,q}$ to restore the full update [64], never amplifies a noisy reference. Here $\hat{\eta}_{a,q}$ is the estimated probability that the reference is wrong, from the judge's calibrated probability for a decision and from the left-out posterior otherwise (with more answers, the gap between the two likeliest answers' probabilities).

Debate messages. Each message m_t is credited by how far it moves the judge toward the decision:

$$\text{credit}_t = \omega_q \left[P_J(D_q \mid m_{1:t}) - P_J(D_q \mid m_{1:t-1}) \right], \tag{6}$$

where $P_J(D_q \mid m_{1:t})$ is the judge's probability of the decision when shown only the first t messages of the logged debate, and $\omega_q = 1-2\hat{\eta}_q$, with $\hat{\eta}_q$ the judge's calibrated probability that D_q is wrong. A message that makes the judge more sure of the decision earns positive credit, and one that makes

it less sure earns negative credit. Each message is thus credited with its effect given the earlier ones [1, 94], and the credits of a debate sum to ω_q times the judge's total move toward the decision.

A persuasive message can move the judge toward a wrong decision, so **messages are credited only in debates the judge settled by re-running tool evidence**. Such a decision is a high-confidence label, close to an RLVR reward, but not an answer key: the tool's data can be out of date, and re-running settles only the disputed steps. By the argument above, on yes-or-no questions a message's expected credit is a positive multiple of how far it moved the judge toward the true answer, provided such decisions are right more often than wrong and whether one is right does not depend on that move. A decision settled by argument alone is not used as a label: after such a debate, answers are scored against the other groups' answers, and messages earn no credit.

For both rewards, decisions involving agents already flagged are dropped, planted cases never enter training (R5), and no agent is trained on its own rulings or shares a reward with an agent that judges it [83]. Task 2.2's random human audits estimate each topic's error rate, and training on a topic stops once the audits show that rate above a preset threshold below one half, as both need.

Preliminary results. In a pilot training a 1.5B-parameter model on GSM8K, with three of the eight agents whose answers form the reference sharing a flawed hint, rewarding agreement with the model's own majority answers raised held-out accuracy and then collapsed it, while the reward of Eq. (5) raised it from 55% to 73% without collapse.

Evaluation. On all three benchmarks, we train the same models with our rewards, with rewards for majority agreement and for debate wins [6], and, as an upper bound, on the known answers (standard RLVR); otherwise the known answers are used only for evaluation. We compare the accuracy of the agents' answers and of the collective decisions on held-out questions. The known answers also show, by topic, how often each reference is wrong (tool-settled decisions, other groups' answers), testing the premise $\eta_q < \frac{1}{2}$ and the conditions for message credit directly. Our hypothesis, H5, tested by month 36, is that our rewards gain more than majority-agreement and debate-win rewards and recover at least half the gain of training on the known answers.

2.5 Shared Evaluation Setup and Expected Results

Each task states its own evaluation and hypothesis (H1–H5); this section gives what the tasks share.

Agent pools and red team. Agents are built on open-weight models of several families and sizes, plus commercial models where licenses allow, with varied role instructions and tools. Our red team, modeled on AI control evaluations [30], builds attackers of increasing strength: (1) a fixed dishonest agent; (2) one that knows the protocol but not its random draws; (3) one trained to spot planted debates that cheats rarely, and only when it expects no audit; and (4) a debater and a judge that coordinate through hidden signals. We report the wrong decisions that get through at each audit budget against the useful work still done, as safety–usefulness curves.

Conditions, baselines, and statistics. Experiments vary how widely mistakes are shared (from none to most agents given the same flawed hint or outdated report, across one to five model families), the share of dishonest agents (up to 30%, at red-team levels 1–4), the share of disputed steps a tool can settle and of reports that are outdated or forged, the budgets for debates, audits, and computation (including small models and few tool calls, as under limited connectivity), and the answer type. Regimes where the propositions say a method fails are included, so that failures are measured; transfer is measured on held-out topics and model families. Baselines at matched compute are a single AI judge (with and without the debaters' tools), majority vote, open discussion, and equal-weight averaging; ablations remove one component at a time, and each term of Eq. (1) is

measured separately, so a gain can be traced to the task that produced it. Benchmark B also reports the running example's measure, missed and false departure calls per 100 vessel questions at each audit budget. We report accuracy, calibration (Brier score, expected calibration error), and means with confidence intervals over seeds and questions.

Results and new knowledge. If successful, the project will deliver the methods of Section 1.3 with their theory (identifiability without an answer key, the price of impartiality, bounds on coalition gains and on a colluder's harm, and rewards that move agents toward correct answers on average), an open debate harness, three benchmarks, and red-team protocols. Scientifically, it will show when a debate among AI agents can be trusted and how few audits keep the agents honest, which would let conferences and funders try impartial, audited peer review by AI; because people audit only a small share of decisions and training needs no labels, the methods can scale to AI whose answers people cannot easily evaluate.

2.6 Project Schedule, Milestones, Goals, and Benchmarks

The project runs 36 months from April 1, 2027, with milestones M1–M6 (Table 2).

Goals and benchmarks. The research goals are RG1–RG3 (Section 1.2). Progress toward them is measured on benchmarks A–C (Table 1) and PeerEval against the success criteria of hypotheses H1–H5, each stated with its target in its task's evaluation (Sections 2.2–2.4). Targets sit near or below the pilot results because they must hold on harder benchmarks and against stronger red teams than the pilots faced. H1–H3 are tested at M4 (month 24) and H4–H5 at M6 (month 36), with interim results at other milestones and the annual Program Reviews; misses are reported as findings. Code, benchmarks, protocols, and peer-reviewed papers with proofs are released by month 36.

Activity	Y1a	Y1b	Y2a	Y2b	Y3a	Y3b
Benchmarks and red team (Sections 2.1 and 2.5)						
Benchmarks A and B; red-team levels 1–2						
Benchmark C; red-team level 3						
Red-team level 4; full evaluation; release						
RG1: Weigh answers, and choose the debaters and the judge						
Task 1.1: reliability model, identifiability; then topics, error costs						
Task 1.2: impartial choice; then coalitions, price of impartiality						
RG2: Settle disputes by agentic debate, and audit for collusion						
Task 2.1: harness and judges; then sub-step recursion, right of reply						
Task 2.2: planted and human audits; then audit bounds						
RG3: Self-improve with reinforcement learning on collective decisions						
Task 3: rewards (5)–(6); then comparison						

Milestones. M1 (month 6): harness v1 and benchmark A released. M2 (12): reliability model and its identifiability analysis, on benchmarks A and B. M3 (18): impartial choice of debaters and judges integrated; tool-backed debate on benchmark B. M4 (24): audit scheme against red-team levels 1–3; benchmark C; H1–H3 tested. M5 (30): audit bounds; analysis and comparison of rewards. M6 (36): full framework against level 4; H4–H5 tested; code, benchmarks, and protocols released.

Table 2: Schedule by half-year (Y1 = April 2027 to March 2028) and milestones.

2.7 Gold Standard Science

The PI's group already follows the tenets of Executive Order 14303: MEQA, DSBench, and LMR-Bench were released with their data and code [41, 55, 99], and OpenAI has since used DSBench to evaluate ChatGPT agent [74]. This project continues that practice: every debate is logged with exact tool calls, prompts, and seeds for re-running; code, benchmarks, and protocols are released with each paper; guarantees are stated with their assumptions; hypotheses H1–H5 and analysis plans are fixed in advance; a red team attacks our own methods; and negative results are published.

3 Management Approach

This is a single-PI award with no subawardees, collaborators, co-investigators, or senior personnel. The PI directs all research, submits annual and final reports, attends ONR Program Reviews, and publishes in peer-reviewed venues, releasing code, benchmarks, and protocols at M1, M6, and with each paper. The budget supports two Ph.D. students for three years, one month of PI summer effort per year, travel to ONR Project and Program Review Meetings and to conferences, and cloud computing and model access.

Student 1 (statistics and mechanism design) leads Tasks 1.1 and 1.2, the audit analysis of Task 2.2, and benchmarks A and C. Student 2 (LLM agents and reinforcement learning) leads the harness and trained judges of Task 2.1, the planted-case generator of Task 2.2, Task 3, and benchmark B. Both run the red team, rotating roles so that no one attacks only their own method. The PI meets each student weekly and reviews progress against the milestones (Table 2) and the success criteria of H1–H5 every six months.

4 Principal Investigator Qualifications

The PI, Xinya Du, has been an Assistant Professor of Computer Science at The University of Texas at Dallas since 2022, after a Ph.D. in Computer Science from Cornell University (2021, advised by Claire Cardie) and postdoctoral research with Heng Ji at the University of Illinois Urbana-Champaign. The PI has 66 papers and preprints, including 28 in the main proceedings of NeurIPS, ICLR, ICML, ACL, EMNLP, NAACL, EACL, and AAAI, cited 4,735 times (h-index 29, Google Scholar, October 2026); $1.02M in external research funding as sole PI; and three Ph.D. students, two finishing in 2026, with former mentees now in Ph.D. programs at CMU, Cornell, and NTU. The most pertinent contributions, most of them made with students at UT Dallas, are these:

Peer evaluation among AI judges. PRD [54] introduced peer ranking and peer discussion among AI judges, which agree with human judgments better than the strongest single judge and reduce a judge's bias toward its own model; researchers at NYU, Anthropic, and Cohere have built on it. It underlies PeerEval (peereval-ai.github.io), a public leaderboard on which seven AI models answer the same questions and judge one another anonymously (40,156 pairwise judgments in its first release). IQA-EVAL [56] lets AI agents with assigned personas play the users of an interactive question-answering system and then evaluate the interactions, agreeing closely with human evaluators. PRD and PeerEval are the basis for Task 1.2's weighted peer ratings and its preliminary results, and IQA-EVAL supports giving the agents of Section 2.5 varied role instructions.

Faithfulness evaluation across modalities. FaithScore [40] splits a vision-language model's free-form answer into atomic facts and checks each against the image; groups at Arizona State, the University of Washington, Johns Hopkins, and Intel Labs have adopted it. FIFA [42] extends such fact-level checks to text-to-video and video-to-text generation, agreeing with human judgment

more closely than earlier methods. FIHA [98] evaluates hallucinations in vision-language models automatically with scene graphs, and Dr.V [61] diagnoses video hallucinations by grounding them in space and time. Checking one claim at a time against evidence is also how Task 2.1's judge rules on each disputed step.

Step-level rewards and verification. FG-PRM [57] detects six types of hallucination at each step of a math solution, and HiPRAG [95] rewards each search decision of an agent, cutting needless searches to 2.3% and missing fewer needed ones. HiPRAG underlies Task 3's message credit.

Question generation and benchmarks. A paper from the PI's Ph.D. on neural question generation [22] has been cited more than 1,000 times and is ranked by Paper Digest among the most influential ACL 2017 papers. MEQA [55] composes multi-hop questions with explanations from event structures and builds benchmark B. DSBench [41], which OpenAI used to evaluate ChatGPT agent [74], and LMR-Bench [99] test agents that analyze data and reproduce research code, the kind of tool use that Task 2.1's judge re-runs and benchmark C checks.

Adversarial evaluation. The PI co-advised one of the five university red teams in the Amazon Nova AI Challenge [79, 96], whose findings shape Task 2.2's planted cases, and holds an Open Philanthropy grant on control evaluations of AI debate.

The PI's awards include the NSF CAREER Award (2024–2029, funded on first submission), Amazon (2023) and Cisco (2024) Research Awards, the Best Paper Award of the AAAI 2025 AI4Research Workshop, AAAI New Faculty Highlights (2024), and the University of Chicago Rising Star in Data Science (2021). The PI is an area chair for ICML 2027, ICLR 2027, and NeurIPS 2026, has been one for ICLR 2025, ACL, EMNLP (Outstanding Area Chair Award, 2024), and COLING, and has served on NSF review panels, including for CAREER. As a postdoctoral researcher, the PI led the integration of RESIN, the event-prediction system built under a $10M DARPA KAIROS grant and ranked first in its evaluation, and was first author of RESIN-11 [23].

5 Responsibility

Resources. UT Dallas provides the PI's laboratory space, a GPU server (ten A100 GPUs), and the shared Juno cluster (67 GPUs; see Facilities and Other Resources). With the requested cloud and model-access costs, these suffice; no equipment purchase is requested.

Ability to comply. The UT Dallas Office of Research and Innovation administers the PI's federal awards and will administer this award under 2 CFR 200 and ONR terms; the PI's commitments, listed in Current and Pending Support, leave the proposed effort available.

Performance history. The PI has managed an NSF CAREER award (2024–2029) and industry and foundation awards, filing reports on time; funds on current awards are being spent on schedule.

Integrity and business ethics. The PI and UT Dallas follow federal research-integrity policies (DoDI 3210.7); the PI has no record of misconduct.

Qualifications and eligibility. The PI is a U.S. permanent resident, has held a full-time tenure-track appointment at UT Dallas since 2022, the PI's first, and received the Ph.D. from Cornell University in 2021, on or after January 1, 2019, as Section B of the NOFO requires.

Organization and controls. UT Dallas maintains federally reviewed accounting, property-control, research-security, quality-assurance, and safety systems and is registered in SAM (UEI EJCVPNN1WFS5; CAGE 0W921); the research is computational.

Xinya Du

Curriculum Vitae · October 2026

Contact

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Research Impact

- **Citations.** 4,735 citations, h-index 29, i10-index 41 (Google Scholar, October 2026). "Learning to Ask" (ACL 2017) [51] has over 1,000 citations, is among Paper Digest's most influential ACL papers and the top 0.1% most-cited computer science articles of 2017, and was featured in *New Scientist* and *TechRepublic*; "Event Extraction by Answering (Almost) Natural Questions" (EMNLP 2020) [44] has over 600. Six papers from 2024–2025 already have over 150 citations each [1], [3], [5], [19], [21], [23].
- **Adoption beyond the lab.** OpenAI used DSBench [19] to evaluate ChatGPT agent (2025). PRD's peer evaluation of AI judges [5] has been built on by researchers at NYU, Anthropic, and Cohere; FaithScore [23] has been adopted by groups at Arizona State, the University of Washington, Johns Hopkins, and Intel Labs. Led the LLM usage-and-licensing section of *The Practical Guides for LLMs* on GitHub (10k+ stars).
- **Publications.** 66 papers and preprints: 5 journal articles (ACM Computing Surveys, IJCV, TMLR) and 46 peer-reviewed conference papers, 28 of them in the main proceedings of NeurIPS, ICLR, ICML, ACL, EMNLP, NAACL, EACL, and AAAI.
- **Recognition.** NSF CAREER Award (2024, funded on first submission); Best Paper Award, AAAI 2025 AI4Research Workshop; Best Poster Award (top 1%), ICML 2024 AI for Science Workshop; Outstanding Area Chair Award, EMNLP 2024; AAAI New Faculty Highlights (2024); Amazon (2023) and Cisco (2024) Research Awards; University of Chicago Rising Star in Data Science (2021).
- **Systems and resources.** First author of RESIN-11 [36]; led the integration of RESIN, the event-prediction system built under a $10M DARPA KAIROS grant and the top-ranked system in the program's evaluation. Built the PeerEval public leaderboard on PRD [5], and benchmarks including MEQA, DSBench, LMR-Bench, and SciEvent.
- **Community leadership.** Area Chair for ICML (2027), NeurIPS (2026), ICLR (2025, 2027), ACL, EMNLP, and COLING; NSF review panels, including CAREER (2025, 2026); co-organizer of tutorials at AAAI 2026 and ICMR 2025 and of workshops at AAAI 2025, CVPR 2025, and IJCAI 2025; over 20 invited talks and panels.
- **Funding.** $1.02M in external research funding as sole PI (NSF CAREER, Open Philanthropy, Amazon, Cisco, Adobe, OpenAI), plus Amazon awards as co-PI and lead contributor.
- **Mentoring and teaching.** Three Ph.D. students (two finishing in 2026) and over 20 master's, undergraduate, and K–12 students; former mentees went on to Ph.D. programs at CMU, Cornell, and NTU. An undergraduate mentee published in TMLR and won a Jonsson School award, and K–12 outreach projects led to the paper underlying the NSF CAREER award. New LLM course rated 4.75/5 (instructor) and 4.65/5 (course); President's Teaching Excellence Award nominee.
- **Most related to this proposal** ($\star$ in the publication list). AI judges that evaluate one another (PRD [5], IQA-EVAL [24]); step-level rewards and verification (FG-PRM [15], HiPRAG [10], training-free verification [12]); evaluation benchmarks (MEQA [22], FaithScore [23], LMR-Bench [14]); red-teaming of code models (COMET [57]); and AI for science (LDC [6], LLM4SR [1]).

Research Interests

Natural language processing and large language models: reasoning, post-training, and alignment; trustworthy evaluation and oversight of AI agents (peer evaluation, debate, process reward models); information extraction; AI for science.

Education

Cornell University Aug 2016 – Aug 2021
Ph.D. in Computer Science (M.S. degree granted in Aug 2019)
Dissertation: *Towards More Intelligent Extraction of Information from Documents*
Advisor: Claire Cardie. Worked closely with Sasha Rush.

Shanghai Jiao Tong University Sep 2012 – Aug 2016
B.E. in Computer Science and Engineering
Outstanding Graduates Award

Professional Experience

The University of Texas at Dallas Aug 2022 – Present
Assistant Professor (tenure-track), Department of Computer Science

University of Illinois at Urbana-Champaign Sep 2021 – Aug 2022
Postdoctoral Research Associate. Worked with Heng Ji.

Industrial Experience

Google AI Research May 2020 – Aug 2020
Research Intern. Supervisor: Luheng He, Qi Li, Emily Pitler.

Allen Institute for Artificial Intelligence (AI2) Sep 2018 – Dec 2018
Research Intern. Supervisor: Bhavana Dalvi, Peter Clark.

Microsoft Research (MSR) May 2018 – Aug 2018
Research Intern. Supervisor: Paul Bennett, Ahmed Awadallah.

Awards & Honors

Best Paper Award (for LDC [6]) AAAI AI4Research Workshop, 2025
Faculty Research Award Department of Computer Science, UT Dallas, 2025
NSF CAREER Award National Science Foundation, 2024
Outstanding Area Chair Award EMNLP, 2024
Best Poster Award (top 1% of 250 submissions; [21]) ICML AI for Science Workshop, 2024
AAAI New Faculty Highlights (paper [27]) AAAI, 2024
Cisco Faculty Research Award Cisco, 2024
President's Teaching Excellence Award Nominee UT Dallas, 2024
Amazon Research Award Amazon, 2023
Spotlight Rising Star in Data Science University of Chicago, 2021
Top 100 New Stars in Artificial Intelligence Baidu Scholar, 2020
Most Influential ACL Papers (15 each year; [51]) Paper Digest, 2017
Outstanding Graduates Award Shanghai Jiao Tong University, 2016
National Scholarship (top 1% of students nationwide) SJTU, 2013
National Olympiad in Mathematics in Provinces, First Prize 2011
National Olympiad in Informatics in Provinces, First Prize 2010

Paper-level recognition (oral presentations, top-1% and top-0.1% most-cited designations, media coverage, and the DARPA KAIROS result) is noted with each publication below.

Grants

Control Evaluations for AI Debate (Single PI)
Funding Source: Open Philanthropy.
Period: Oct 2025 – present.
Amount Awarded: $218,513 (+ $40,000 computing). Amount to me: $218,513.

NSF CAREER: Learning to Extract Consistent Event Graphs from Long and Complex Documents (Single PI)
Funding Source: National Science Foundation (NSF).
Period: May 2024 – present.
Amount Awarded: $561,219. Amount to me: $561,219.
Awarded on first submission.

Adobe Research Gift: Hallucinations in Video Language Models (Single PI)
Funding Source: Adobe.
Period: Sep 2025 – present.
Amount Awarded: $30,000.

Process-guided Fine-tuning for Answering Complex Questions (Single PI)
Funding Source: Amazon Research Award.
Period: Jan 2024 – present.
Amount Awarded: $110,000 ($40,000 AWS credits). Amount to me: $110,000.

Amazon Trusted AI Challenge Grant (Co-PI, with Prof. Wei Yang)
Funding Source: Amazon.
Period: Jan 2025 – present.
Amount Awarded to Our Team: $250,000.

Optimizing RISC-V Compilers with RISC-LLM and Syntax Parsing (Lead Contributor; PI: Wei Yang)
Funding Source: Amazon Research Award.
Period: Sep 2025 – Aug 2026.
Amount Awarded: $70,000. Amount to me: $23,000.

FAIGen: Faithful LLM Generation with Scientific Principles-guided Learning (Single PI)
Funding Source: Cisco Faculty Research Award.
Period: Jan 2025 – present.
Amount Awarded: $90,000. Amount to me: $90,000.

Mitigating Hallucinations in Vision-language Models with Fine-grained AI Feedback (Single PI)
Funding Source: OpenAI Researcher Access Program.
Period: Sep 2024 – present.
Amount Awarded: $12,000. Amount to me: $12,000.

Undergraduate Research Apprenticeship Award (URAP) (Single PI)
Funding Source: University of Texas at Dallas.
Period: Summer 2023, 2024, 2025.
Amount Awarded: $13,500. Amount to me: $13,500.

New Faculty Research Symposium Grant (Leading PI)
Funding Source: University of Texas at Dallas.
Period: Aug 2025 – Jul 2026.
Amount Awarded: $25,000.

Grants (Pending)

Collaborative Research: RI: SMALL: Explainable Methods for Natural Language Reasoning (Leading PI, co-PI: Prof. Claire Cardie)
Funding Source: National Science Foundation (NSF).
Period: Jan 2025 – Dec 2028.
Amount Requested: $600,000. Amount to me: $300,000.
Status: Pending (**rated highly competitive on first submission**).

ACED: Large Language Model-based Research Agent for Synthetic Biology Research Idea Generation (Leading PI, co-PI: Prof. Leo Bleris)
Funding Source: National Science Foundation (NSF).
Period: Aug 2026 – Dec 2028.
Amount Requested: $600,000. Amount to me: $300,000.
Status: Pending (**rated competitive on first submission**).

III: Automating Scientific Knowledge Discovery through Retrieval, Generation, and Validation (Leading PI, co-PI: Prof. Vincent Ng)
Funding Source: National Science Foundation (NSF).
Period: Jan 2027 – Dec 2029.
Amount Requested: $600,000. Amount to me: $300,000.
Status: Pending.

LLM-Guided Agents for Rational and Reliable Cell Reprogramming (Co-PI, co-PI: Prof. Leo Bleris)
Funding Source: National Science Foundation (NSF) Directorate for Biological Sciences (BIO).
Period: Aug 2026 – Dec 2029.
Amount Requested: $726,459. Amount to me: $242,153.
Status: Pending.

Publications

Notes: † co-leading author; * equal contribution; ⋆ work most related to the proposed ONR research. Current lists are also on [DBLP] and [Google Scholar]. In computer science, peer-reviewed conference proceedings are the primary archival venue for research; the main tracks of NeurIPS, ICLR, ICML, ACL, EMNLP, NAACL, EACL, and AAAI are highly selective, most accepting roughly 20–30% of submissions.

Journal Articles

[1] ⋆**LLM4SR: A Survey on Large Language Models for Scientific Research** [pdf]
Ziming Luo, Zonglin Yang, Zexin Xu, Wei Yang, **Xinya Du**
In ACM Computing Surveys (CSUR), to appear.

[2] **Dr.V: A Hierarchical Perception-Temporal-Cognition Framework to Diagnose Video Hallucination by Fine-Grained Spatial-Temporal Grounding** [pdf]
Meng Luo, Shengqiong Wu, Liqiang Jing, Tianjie Ju, Li Zheng, Jinxiang Lai, Tianlong Wu, **Xinya Du**, Jian Li, Siyuan Yan, Jiebo Luo, William Yang Wang, Hao Fei, Mong-Li Lee, Wynne Hsu
In International Journal of Computer Vision (IJCV), 2026.

[3] **SFT or RL? An Early Investigation into Training R1-Like Reasoning Large Vision-Language Models** [pdf]
Hardy Chen, Haoqin Tu, Fali Wang, Hui Liu, Xianfeng Tang, **Xinya Du**, Yuyin Zhou, Cihang Xie
In Transactions on Machine Learning Research (TMLR), 2025.

[4] **FGAIF: Aligning Large Vision-Language Models with Fine-grained AI Feedback** [pdf]
Liqiang Jing, **Xinya Du**
In Transactions on Machine Learning Research (TMLR), 2025.

[5] ⋆**PRD: Peer Rank and Discussion Improve Large Language Model-based Evaluations** [pdf]
Ruosen Li, Teerth Patel, **Xinya Du**
In Transactions on Machine Learning Research (TMLR), 2024.
Public leaderboard built on this work: peereval-ai.github.io

Peer-Reviewed Conference Papers (main, Findings, and demonstration tracks)

[6] ⋆**LDC: Learning to Generate Research Idea with Dynamic Control** [pdf]
Ruochen Li, Liqiang Jing, Chi Han, Jiawei Zhou, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.
Earlier version in AAAI 2025 AI4Research Workshop. **(Oral)**
Best Paper Award, AAAI 2025 AI4Research Workshop.

[7] **A Comprehensive Analysis for Visual Object Hallucination in Large Vision-Language Models** [pdf]
Liqiang Jing, Hardy Chen, Ehsan Aghazadeh, Xin Eric Wang, **Xinya Du**
In Findings of the Association for Computational Linguistics: EMNLP, 2026.

[8] **A Skill-augmented Agentic Framework and Benchmark for Multi-Video Understanding** [pdf]
Yue Zhang, Liqiang Jing, Jia Li, Yapeng Tian, **Xinya Du**, Yunhui Guo, Vibhav Giridhar Gogate
In Findings of the Association for Computational Linguistics: EMNLP, 2026.

[9] **AMARIS: A Memory-Augmented Rubric Improvement System for Rubric-Based Reinforcement Learning** [pdf]
Peilin Wu, Xinlu Zhang, Kun Wan, Wentian Zhao, Gang Wu, **Xinya Du**, Zhiyu Chen
In Findings of the Association for Computational Linguistics: EMNLP, 2026.

[10] ⋆**HiPRAG: Hierarchical Process Rewards for Efficient Agentic Retrieval Augmented Generation** [pdf]
Peilin Wu, Mian Zhang, Kun Wan, Wentian Zhao, Kaiyu He, **Xinya Du**, Zhiyu Chen
In International Conference on Learning Representations (ICLR), 2026.

[11] **Is Grokking Worthwhile? Functional Analysis and Transferability of Generalization Circuits in Transformers** [pdf]
Kaiyu He, Mian Zhang, Peilin Wu, **Xinya Du**, Zhiyu Chen
In Findings of the Association for Computational Linguistics: ACL, 2026.

[12] ⋆**Your Reasoning Model is Secretly a Reward Model – Optimization-Free Verification from Experience** [pdf]
Zhenwen Liang, Ruosen Li, Yujun Zhou, Linfeng Song, Dian Yu, **Xinya Du**, Haitao Mi, Dong Yu
In Annual Meeting of the Association for Computational Linguistics (ACL), 2026.

[13] **FIFA: Unified Faithfulness Evaluation Framework for Text-to-Video and Video-to-Text Generation** [pdf]
Liqiang Jing, Viet Dac Lai, Seunghyun Yoon, Trung Bui, **Xinya Du**
In Findings of the Association for Computational Linguistics: ACL, 2026.

[14] ⋆**LMR-Bench: Evaluating LLM Agent's Ability on Reproducing Language Modeling Research** [pdf]
Shuo Yan, Ziming Luo, Zimu Wang, Ruochen Li, Daoyang Li, Liqiang Jing, Kaiyu He, Peilin Wu, Juntong Ni, George Michalopoulos, Yue Zhang, Ziyang Zhang, Mian Zhang, Zhiyu Chen, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[15] ⋆**FG-PRM: Fine-grained Hallucination Detection and Mitigation in Language Model Mathematical Reasoning** [pdf]
Ruosen Li, Ziming Luo, **Xinya Du**
In Findings of the Association for Computational Linguistics: EMNLP, 2025.

[16] **SciEvent: Benchmarking Multi-domain Scientific Event Extraction** [pdf]
Bofu Dong, Pritesh Shah, Sumedh Sonawane, Tiyasha Banerjee, Erin Brady, **Xinya Du**, Ming Jiang
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[17] **Search Wisely: Mitigating Sub-optimal Agentic Searches By Reducing Uncertainty** [pdf]
Peilin Wu, Mian Zhang, Xinlu Zhang, **Xinya Du**, Zhiyu Chen
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[18] **VistaDPO: Video Hierarchical Spatial-Temporal Direct Preference Optimization for Large Video Models** [pdf]
Haojian Huang*, Haodong Chen*, Shengqiong Wu, Meng Luo, Jinlan Fu, **Xinya Du**, Hanwang Zhang, Hao Fei
In International Conference on Machine Learning (ICML), 2025.

[19] **DSBench: How Far Are Data Science Agents from Becoming Data Science Experts?** [pdf]
Liqiang Jing, Zhehui Huang, Xiaoyang Wang, Wenlin Yao, Wenhao Yu, Kaixin Ma, Hongming Zhang, **Xinya Du**, Dong Yu
In International Conference on Learning Representations (ICLR), 2025.
Used by OpenAI to evaluate ChatGPT agent. [Link]

[20] **FIHA: Autonomous Hallucination Evaluation in Vision-Language Models with Davidson Scene Graphs** [pdf]
Bowen Yan, Zhengsong Zhang, Liqiang Jing, Eftekhar Hossain and **Xinya Du**
In Findings of the Association for Computational Linguistics: ACL, 2025.

[21] **Large Language Models for Automated Open-domain Scientific Hypotheses Discovery** [pdf]
Zonglin Yang, **Xinya Du**†, Junxian Li, Jie Zheng, Soujanya Poria, Erik Cambria
In Findings of the Association for Computational Linguistics: ACL, 2024.
Best poster award in ICML AI4Science, 2024 (Top 1% of 250 submissions).

[22] ⋆**Multi-hop Event-centric Question Answering with Explanations**
Ruosen Li, Zimu Wang, Son Quoc Tran, Lei Xia, **Xinya Du**
In Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.

[23] ⋆**FaithScore: Fine-grained Evaluations of Hallucinations in Large Vision-Language Models** [pdf]
Liqiang Jing, Ruosen Li, Yunmo Chen, **Xinya Du**
In Findings of the Association for Computational Linguistics: EMNLP, 2024.

[24] ⋆**IQA-EVAL: Automatic Evaluation of Human-Model Interactive Question Answering** [pdf]
Ruosen Li, Ruochen Li, Barry Wang, **Xinya Du**
In Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.

[25] **QAEvent: Event Extraction as Question-Answer Pairs Generation** [pdf]
Milind Choudhary, **Xinya Du**
In Findings of the Association for Computational Linguistics: EACL, 2024.

[26] **Language Models as Inductive Reasoners** [pdf]
Zonglin Yang, Li Dong, **Xinya Du**†, Hao Cheng, Erik Cambria, Xiaodong Liu, Jianfeng Gao, Furu Wei
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2024. **(Oral)**

[27] **Making Natural Language Reasoning Explainable and Faithful** [pdf]
Xinya Du
In AAAI Conference on Artificial Intelligence (AAAI), 2024. New Faculty Highlights. **(Oral)**

[28] **Proto-CLIP: Vision-Language Prototypical Network for Few-Shot Learning** [pdf]
Jishnu Jaykumar P, Kamalesh Palanisamy, Yu-Wei Chao, **Xinya Du**, Yu Xiang
In International Conference on Intelligent Robots and Systems (IROS), 2024. **(Oral)**

[29] **Document-level Causal Relation Extraction with Knowledge-guided Binary Question Answering** [pdf]
Zimu Wang, Lei Xia, Wei Wang, **Xinya Du**
In Findings of the Association for Computational Linguistics: EMNLP, 2024.

[30] **Leveraging Structured Information for Explainable Multi-hop Question Answering and Reasoning** [pdf]
Ruosen Li, **Xinya Du**
In Findings of the Association for Computational Linguistics: EMNLP, 2023.

[31] **Process of Elimination for Multiple Choice Reasoning** [pdf]
Chenkai Ma, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023.

[32] **Probing Representations for Document-level Event Extraction** [pdf]
Barry Wang, **Xinya Du**, Claire Cardie
In Findings of the Association for Computational Linguistics: EMNLP, 2023.

[33] **End-to-end Case-Based Reasoning for Commonsense Knowledge Base Completion** [pdf]
Zonglin Yang, **Xinya Du**†, Erik Cambria, Claire Cardie
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2023. **(Oral)**

[34] **Zero-Shot Classification by Logical Reasoning on Natural Language Explanations** [pdf]
Chi Han, Hengzhi Pei, **Xinya Du**, Heng Ji
In Findings of the Association for Computational Linguistics: ACL, 2023.

[35] **Logical Entity Representation in Knowledge-Graphs for Differentiable Rule Learning** [pdf]
Chi Han, Qizheng He, Charles Yu, **Xinya Du**, Hanghang Tong, Heng Ji
In International Conference on Learning Representations (ICLR), 2023.

[36] **RESIN-11: Schema-guided Event Prediction for 11 Newsworthy Scenarios** [pdf]
Xinya Du, Zixuan Zhang, Sha Li, Heng Ji, the RESIN team
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): System Demonstrations, 2022.
Top ranking system in DARPA KAIROS evaluation. [Link]

[37] **Retrieval-Augmented Generative Question Answering for Event Argument Extraction** [pdf]
Xinya Du, Heng Ji
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2022.

[38] **Dynamic Global Memory for Document-level Argument Extraction** [pdf]
Xinya Du, Sha Li, Heng Ji
In Annual Meeting of the Association for Computational Linguistics (ACL), 2022.

[39] **Automatic Error Analysis for Document-level Information Extraction** [pdf]
Aliva Das*, **Xinya Du***, Barry Wang*, Kejian Shi, Jiayuan Gu, Thomas Porter, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2022.

[40] **Template Filling with Generative Transformers** [pdf]
Xinya Du, Alexander M. Rush, Claire Cardie
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2021.

[41] **GRIT: Generative Role-filler Transformers for Document-level Event Entity Extraction** [pdf]
Xinya Du, Alexander M. Rush, Claire Cardie
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2021. **(Oral)**

[42] **Few-shot Intent Classification and Slot Filling with Retrieved Examples** [pdf]
Dian Yu, Luheng He, Yuan Zhang, **Xinya Du**, Panupong Pasupat, Qi Li
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2021. **(Oral)**

[43] **QA-Driven Zero-shot Slot Filling with Weak Supervision Pretraining** [pdf]
Xinya Du, Luheng He, Qi Li, Dian Yu, Panupong Pasupat, Yuan Zhang
In Annual Meeting of the Association for Computational Linguistics (ACL), 2021.

[44] **Event Extraction by Answering (Almost) Natural Questions** [pdf]
Xinya Du, Claire Cardie
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020. **(Oral)**
Top 1% most cited articles published in Computer Science in 2020.

[45] **Improving Event Duration Prediction via Time-aware Pre-training** [pdf]
Zonglin Yang, **Xinya Du**, Alexander M. Rush, Claire Cardie
In Findings of the Association for Computational Linguistics: EMNLP, 2020.

[46] **Document-Level Event Role Filler Extraction using Multi-Granularity Contextualized Encoding** [pdf]
Xinya Du, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2020.

[47] **Leveraging Structured Metadata for Improving Question Answering on the Web** [pdf]
Xinya Du, Adam Fourney, Robert Sim, Claire Cardie, Paul Bennett, Ahmed Hassan Awadallah
In Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL/IJCNLP), 2020.

[48] **Be Consistent! Improving Procedural Text Comprehension using Label Consistency** [pdf]
Xinya Du, Bhavana Dalvi, Niket Tandon, Antoine Bosselut, Wen-tau Yih, Peter Clark, Claire Cardie
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2019.

[49] **Harvesting Paragraph-Level Question-Answer Pairs from Wikipedia** [pdf]
Xinya Du, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2018. **(Oral)**
Top 1% most cited articles published in Computer Science in 2018.

[50] **Identifying Where to Focus in Reading Comprehension for Neural Question Generation** [pdf]
Xinya Du, Claire Cardie
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2017.

[51] **Learning to Ask: Neural Question Generation for Reading Comprehension** [pdf]
Xinya Du, Junru Shao, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2017.
Most Influential ACL Papers (15 each year). [Link]
Featured in New Scientist and TechRepublic. [Link]
Top 0.1% most cited articles published in Computer Science in 2017.

Workshop Papers, Tutorial Abstracts, and Shared-Task Reports

[52] **Efficient Document-level Event Relation Extraction**
Ruochen Li, Zimu Wang, **Xinya Du**
In 10th Workshop on Representation Learning for NLP at Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2025.

[53] **Hallucinations in Large Language Models and Large Vision-Language Models (Tutorial)** [pdf]
Liqiang Jing, Yue Zhang, **Xinya Du**
In Proceedings of the ACM International Conference on Multimedia Retrieval (ICMR), pp. 2138–2139, 2025.

[54] **Logical Reasoning over Natural Language as Knowledge Representation: A Survey** [pdf]
Zonglin Yang, **Xinya Du**, Rui Mao, Jinjie Ni, Erik Cambria
In Workshop on Natural Language Reasoning and Structured Explanations (NLRSE) at Annual Meeting of the Association for Computational Linguistics (ACL), 2023.

[55] **Toward Consistent and Informative Event-Event Temporal Relation Extraction** [pdf]
Xiaomeng Jin, Haoyang Wen, **Xinya Du**, Heng Ji
In MATCHING at Annual Meeting of the Association for Computational Linguistics (ACL), 2023. **(Oral)**

[56] **Cornell Belief and Sentiment System at TAC 2016** [pdf]
Vlad Niculae, Kai Sun, Xilun Chen, Yao Cheng, **Xinya Du**, Esin Durmus, Arzoo Katiyar, Claire Cardie
In Text Analysis Conference (TAC), 2016.

Challenge Proceedings

[57] ⋆**COMET: Closed-loop Orchestration for Malicious Elicitation Techniques in Code Models** [pdf]
Z. Xu, T. Li, R. S. Rathnasuriya, Z. Song, J. Ren, B. Mandalapu, S. Setayeshpour, **Xinya Du**, Wei Yang
In 1st Proceedings of Amazon Nova AI Challenge (Trusted AI), 2025.

Preprints

[58] **Be Consistent! Enhancing Robust Visual Reasoning in LVLMs with Consistency Constraints** [pdf]
Liqiang Jing, Xiong Zhou, Siddharth Varia, Neha Anna John, **Xinya Du**, Vassilis N. Ioannidis
arXiv:2607.21722, 2026.

[59] **Event Extraction in Large Language Model** [pdf]
Bobo Li, Xudong Han, Jiang Liu, Yuzhe Ding, Liqiang Jing, Zhaoqi Zhang, Jinheng Li, **Xinya Du**, Fei Li, Meishan Zhang, Min Zhang, Aixin Sun, Philip S. Yu, Hao Fei
arXiv:2512.19537, 2025.

[60] **GEAR: A General Evaluation Framework for Abductive Reasoning** [pdf]
Kaiyu He, Peilin Wu, Mian Zhang, Kun Wan, Wentian Zhao, **Xinya Du**, Zhiyu Chen
arXiv:2509.24096, 2025.

[61] **AALC: Large Language Model Efficient Reasoning via Adaptive Accuracy-Length Control** [pdf]
Ruosen Li, Ziming Luo, Quan Zhang, Ruochen Li, Ben Zhou, Ali Payani, **Xinya Du**
arXiv:2506.20160, 2025.

[62] **Multimodal Reference Visual Grounding** [pdf]
Yangxiao Lu, Ruosen Li, Liqiang Jing, Jikai Wang, **Xinya Du**, Yunhui Guo, Nicholas Ruozzi, Yu Xiang
arXiv:2504.02876, 2025.

[63] **Do Retrieval-Augmented Language Models Adapt to Varying User Needs?** [pdf]
Peilin Wu, Xinlu Zhang, Wenhao Yu, Xingyu Liu, **Xinya Du**, Zhiyu Zoey Chen
arXiv:2502.19779, 2025.

[64] **MLR-Copilot: Autonomous Machine Learning Research based on Large Language Models Agents** [pdf]
Ruochen Li, Teerth Patel, Qingyun Wang, **Xinya Du**
arXiv:2408.14033, 2024.
The first autonomous LLM agent for the entire machine learning research process: generating hypotheses, implementing ideas, and running experiments.

[65] **VIA: Unified Spatiotemporal Video Adaptation Framework for Global and Local Video Editing** [pdf]
Jing Gu, Yuwei Fang, Ivan Skorokhodov, Peter Wonka, **Xinya Du**, Sergey Tulyakov, Xin Eric Wang
arXiv:2406.12831, 2024.

[66] **AGent: A Novel Pipeline for Automatically Creating Unanswerable Questions** [pdf]
Son Quoc Tran, Gia-Huy Do, Phong Nguyen-Thuan Do, Matt Kretchmar, **Xinya Du**
arXiv:2309.05103, 2023.

Dissertation

[67] **Towards More Intelligent Extraction of Information from Documents**
Xinya Du
Ph.D. Dissertation, Cornell University, 2021.

Professional Services

Grant-Proposal Review Panels

National Science Foundation (NSF) review panel, IIS/RI CAREER program, 2026.

National Science Foundation (NSF) review panel, IIS/RI Future CORe program, 2026.

National Science Foundation (NSF) review panel, IIS/RI MEDIUM program, 2025.

Area Chair Roles

ICML (International Conference on Machine Learning) Area Chair, 2027.

NeurIPS (Conference on Neural Information Processing Systems) Area Chair, 2026.

ICLR (International Conference on Learning Representations) Area Chair, 2025, 2027.

ACL (Annual Meeting of the Association for Computational Linguistics) Area Chair, 2023, 2025; ACL Rolling Review Area Chair, 2024; ACL Demo Track Area Chair, 2026.

EMNLP (Conference on Empirical Methods in Natural Language Processing) Area Chair, 2024 **(Outstanding Area Chair Award)**, 2025; EMNLP Demo Track Area Chair, 2024.

COLING (International Conference on Computational Linguistics) Area Chair, LREC-COLING 2024 and COLING 2025.

Conference Organizing Roles

NAACL (Conference of the North American Chapter of the Association for Computational Linguistics) Website Chair, 2024.

ACL Session Chair, 2024.

Tutorials and Workshops Organized

Frontiers of LLMs for Automated Scientific Discovery: A Tutorial
Organizing team: Zonglin Yang, Chandan Reddy, Xinya Du and Mengdi Wang.
In AAAI Conference on Artificial Intelligence (AAAI), 2026.

Hallucinations in Large Language Models and Large Vision-Language Models: A Tutorial
Organizing team: Liqiang Jing, Yue Zhang, Xinya Du.
In ACM International Conference on Multimedia Retrieval (ICMR), 2025.

AI4Research: Towards a Unified Knowledge-grounded Scientific Research Lifecycle (workshop).
Organizing team: Qingyun Wang, Wenpeng Yin, Lifu Huang, Yi R. Fung, Xinya Du, Carl Edwards, Tom Hope.
In AAAI Conference on Artificial Intelligence (AAAI), 2025.

Evaluations and Benchmarks in Context of Multimodal LLM (workshop).
Organizing team: Hao Fei, Xiang Yue, Kaipeng Zhang, Long Chen, Jian Li, Xinya Du.
In The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

The First Workshop on Multimodal Knowledge and Language Modeling
Organizing team: Liqiang Jing, Xinya Du, Hao Fei, Jing Gu, Manling Li, Aixin Sun, William Wang
In International Joint Conference on Artificial Intelligence (IJCAI), 2025.

Program Committee Member / Reviewer

ACL (Annual Meeting of the Association for Computational Linguistics).

NeurIPS (Conference on Neural Information Processing Systems).

ICLR (International Conference on Learning Representations).

AISTATS (International Conference on Artificial Intelligence and Statistics).

EMNLP (Conference on Empirical Methods in Natural Language Processing).

NAACL (Conference of the North American Chapter of the Association for Computational Linguistics).

AACL/IJCNLP (Asia-Pacific Chapter of the Association for Computational Linguistics / International Joint Conference on Natural Language Processing).

IJCAI (International Joint Conference on Artificial Intelligence).

AAAI (AAAI Conference on Artificial Intelligence).

NLPCC (Natural Language Processing and Chinese Computing).

CoNLL (Conference on Computational Natural Language Learning).

WNUT (Workshop on Noisy User-generated Text).

MRQA (Workshop on Machine Reading for Question Answering).

SEM (Joint Conference on Lexical and Computational Semantics).

Journal Reviewer

TKDE (IEEE Transactions on Knowledge and Data Engineering).

TASLP (IEEE Transactions on Audio, Speech and Language Processing).

TNNLS (IEEE Transactions on Neural Networks and Learning Systems).

TALLIP (ACM Transactions on Asian and Low-Resource Language Information Processing).

TKDD (ACM Transactions on Knowledge Discovery from Data).

CL (Computational Linguistics).

KAIS (Knowledge and Information Systems).

AI Communication.

IPM (Information Processing and Management).

Ph.D./M.S. Thesis Committee Member

Ph.D.: Kangshuo Li, Souradeep Nanda, Ali Riahi, Xiangci Li, Jishnu Jaykumar Padalunkal, Basel Abdeen, Yibo Hu.

M.S.: Wooseong Yang, Shubham Patel.

Department, University, and Outreach Service

Faculty advisor and judge: Association for Computing Machinery (ACM) Symposium, UT Dallas, 2023–present.

Faculty Advisor: Women Who Compute (WWC), UT Dallas, 2024–present.

Faculty Advisor: Society of Asian Scientists and Engineers (SASE), UT Dallas, 2024–present.

Invited Speaker: UTD AI Society's "AI Chats", 2025.

Panelist: FuturED 2024 Workshop on Future of Event Detection, 2024.

Faculty Host: K-12 Outreach, UT Dallas, Summer 2023, 2024.

Research Mentor: Research, Inquiry, Design Experience (RIDE) Project, UT Dallas, 2024.

PhD Admission Committee Member, UT Dallas CS Department, 2022, 2023, 2024.

Site Host: North American Computational Linguistics Olympiad (NACLO), 2023, 2024.

PhD Admission Committee Member, Cornell CS Department, 2021.

Volunteer: Cornell CS Department PhD Visit Day, 2019, 2020, 2021.

Student Volunteer: ACL, 2017, 2018.

Student Volunteer: EMNLP, 2017.

Invited Talks and Panels

Pushing the Boundaries of LLM Agents in Data Science and Machine Learning Research.
Nvidia, Dec 2025.

Inside LLMs: Knowledge, Alignment, and Agents.
Samsung Research America (SRA), May 2025.

Towards Autonomous Scientific Discovery with Large Language Models.
UT Southwestern AI in Medicine Seminar Series, May 2025.

Synergizing Knowledge and Large Language Models.
University of Illinois Urbana-Champaign, Data Mining Group Seminar, Oct 2024.
University of Massachusetts–Amherst, Machine Learning & Friends Lunch, Sep 2024.

Synergy between Large Language Model and Knowledge.
University of North Texas, Oct 2024.

Panelist, Future of Event Detection Workshop.
Conference on Empirical Methods in Natural Language Processing (EMNLP), Nov 2024.

Faculty Round Table Talk.
UT Dallas Hobson Wildenthal Honors College, Aug 2024.

Large Language Models: Knowledge, Reasoning and Factuality.
Samsung Electronics America, Mar 2024.

Making Natural Language Reasoning Explainable and Faithful (New Faculty Highlights).
AAAI Conference on Artificial Intelligence (AAAI), Feb 2024.

Open-ended Evaluations of Foundational Models: Alignment and Faithfulness.
Shanghai Jiao Tong University, Computer Science Department Seminar, Dec 2023.
Fudan University, Computer Science Department Seminar, Dec 2023.

Panelist, "ChatGPT: Fact vs. Fiction" public forum, sponsored by The Dallas Morning News.
UT Dallas, Mar 2023.

Towards More Intelligent Extraction of Information from Documents.
Allen Institute for Artificial Intelligence, Mar 2022.
Facebook, Mar 2022.
Simon Fraser University, Feb 2022.
Rensselaer Polytechnic Institute, Feb 2022.
University of California, Merced, Feb 2022.
UIUC, Siebel School of Computing and Data Science Speaker Series, Feb 2022.

Towards More Informed Extraction of Events from Documents.
University of Chicago, Rising Stars in Data Science Workshop, Jan 2021.
Tencent AI Research America, Nov 2020.

Event Extraction by Answering (Almost) Natural Question.
UIUC, Information Extraction and Knowledge Acquisition Class, Sep 2020.

LwLL: Progress on the NLP Front.
Cornell University, DARPA site visit, Apr 2020.

Mentoring Experience

PhD students

Ruosen Li (2022–2026, UT Dallas PhD student, graduated Aug 2026).
Publications: ACL 2026; NeurIPS 2024 (two papers); TMLR 2024; EMNLP Findings 2023, 2024, 2025.

Liqiang Jing (2023–2026, UT Dallas PhD student, graduating Nov 2026).
Publications: ICLR 2025; EMNLP 2025, 2026; TMLR 2025; IJCV 2026; EMNLP Findings 2024, 2026 (two papers); ACL Findings 2025, 2026; ICMR 2025 tutorial.
Award: OpenAI Researcher Access Program.

Ruochen Li (2023–present, UT Dallas PhD student).
Publications: EMNLP 2026 (first author; an earlier version won the Best Paper Award at the AAAI 2025 AI4Research Workshop); EMNLP 2025; NeurIPS 2024; RepL4NLP Workshop at NAACL 2025.

Master Students

Milind Choudhary (2023, UT Dallas Master student) → UT Dallas PhD.
Topic: Event Extraction as Question-Answer Pairs Generation.
Publications: EACL Findings 2024.

Zimu Wang (2023–present, UT Dallas Visiting student) → UoL PhD.
Topic: Temporal relation extraction.
Publications: NeurIPS 2024, EMNLP 2025, EMNLP Findings 2024, RepL4NLP Workshop at NAACL 2025.

Son Tran (2023, UT Dallas Visiting student) → Cornell PhD.
Topic: Complex Question Answering.
Publications: NeurIPS 2024.

Chenkai Ma (2023, UT Dallas Visiting student).
Topic: Multiple Choice Questions Reasoning.
Publications: EMNLP 2023.

Barry Wang (2021–2024, Cornell CS undergraduate student) → CMU PhD.
Topic: Automatic Error Analysis for Information Extraction.
Publications: ACL 2022, SciNLP 2022, EMNLP Findings 2023, NeurIPS 2024.

Zonglin Yang (2020–2024, Cornell CS MEng student) → NTU PhD.
Topic: Commonsense and Case-based Reasoning for NLP.
Publications: EMNLP Findings 2020, EACL 2023, EACL 2024, ACL Findings 2024 (Best Poster Award, ICML 2024 AI for Science Workshop).

Ehsan Aghazadeh (2024, UT Dallas Visiting student).
Topic: Large Vision Language Models Interpretability.
Publications: EMNLP Findings 2026.

Bowen Yan (2024, UT Dallas Visiting student).
Topic: Large Vision Language Models Hallucinations.
Publications: ACL Findings 2025.

Zhengsong Zhang (2024, UT Dallas Visiting student).
Topic: Large Vision Language Models Hallucinations.
Publications: ACL Findings 2025.

Ziming Luo (2024, UT Dallas Visiting student).
Topic: Large Language Model for Scientific Discovery.
Publications: EMNLP 2025, EMNLP Findings 2025.

Undergraduate/High School Students

Teerth Patel (UT Dallas BS student, 2023–present).
Topic: Large Language Models Peer Evaluations.
Publications: TMLR 2024.
Award: Jonsson School of Engineering and Computer Science Award.

Lei Xia (2023–2024, UT Dallas Visiting student).
Topic: Multi-hop Question Answering.
Publications: NeurIPS 2024, EMNLP Findings 2024.

Minhao Zou (2024, UT Dallas Visiting student).
Topic: LLM for Scientific Discovery.

Arjun Junghare (2024, UT Dallas CS undergraduate student).
Topic: Image Generation with Large Language Models.

Shreya Kumara (2024, President of UTD Society of Asian Scientists and Engineers).

Anjana Bharadwaj (2024, President of UTD Women Who Compute).

Jaden Nunes (Summer 2023, DFW Local K-12 student).
Topic: Event Extraction as Question-Answer Pairs Generation.

Rishab Bhattacharya (Summer 2023, DFW Local K-12 student).
Topic: Event Extraction as Question-Answer Pairs Generation.

Shreyas Kumar (Summer 2023, DFW Local K-12 student).
Topic: Event Extraction as Question-Answer Pairs Generation.

Aliva Das (2021–2022, Cornell CS undergraduate student) → Amazon.
Topic: Automatic Error Analysis for Information Extraction.
Publications: ACL 2022, SciNLP 2021.

Rishi Malhotra (Spring 2021, Cornell CS undergraduate student) → Microsoft.
Topic: Applying Neural Document-level IE Model to the Scientific Domain.

Maitreyi Chatterjee (Spring 2021, Cornell CS undergraduate student) → LinkedIn.
Topic: Applying Neural Document-level IE Model to the Scientific Domain.

Teaching Experience

Introduction to Machine Learning, UT Dallas, Spring 2024.

Natural Language Processing, UT Dallas, Fall 2023, 2024, 2025.

Deep Learning for Natural Language Processing, UT Dallas, Spring 2023.
New course developed and taught (overall instructor score 4.75/5.0; course score 4.65/5.0). [Link]

Natural Language Processing, UT Dallas, Fall 2022.

Natural Language Processing, Cornell University, Fall 2019.
Teaching Assistant for Prof. Claire Cardie.

Natural Language Processing, Cornell University, Spring 2019.
Teaching Assistant for Prof. Yoav Artzi.

Software Engineering, Cornell University, Spring 2017, 2018.
Teaching Assistant for Prof. William Arms.

Introduction to Computing Using Python, Cornell University, Fall 2016.
Teaching Assistant for Prof. Walker White.

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COMPUTER SCIENCE DEPARTMENT

ERIK JONSSON SCHOOL OF
ENGINEERING AND COMPUTER SCIENCE

October 20, 2026

Dear ONR Young Investigator Program Selection Committee:

I am pleased to provide my strongest endorsement for Dr. Xinya Du's application to the Office of Naval Research Young Investigator Program, entitled "Trustworthy Collective Reasoning and Evaluation for AI Agents." As an Assistant Professor in the Department of Computer Science at The University of Texas at Dallas, Dr. Du is an outstanding early-career researcher with exceptional potential to become a leader in artificial intelligence, with a focus on language model reasoning, evaluation, and autonomous agents. Our department is strongly committed to supporting his research and long-term academic career.

Dr. Du was recruited to UT Dallas based on his strong research accomplishments, technical expertise, and ambitious vision for advancing artificial intelligence. His recruitment reflects our department's commitment to attracting outstanding researchers with the potential to make significant contributions to their fields. Since joining the department, he has established an active research program addressing fundamental challenges in the reliability and reasoning capabilities of modern AI systems. His research has resulted in publications at leading conferences in natural language processing and machine learning, including NeurIPS, ACL and EMNLP. His ability to identify important emerging problems and develop original technical solutions distinguishes him as a highly promising young investigator.

Dr. Du's proposed ONR research, "Trustworthy Collective Reasoning and Evaluation for AI Agents," addresses a fundamental challenge in the development of reliable autonomous AI systems: enabling multiple agents to evaluate one another's reasoning, recognize relevant expertise, and improve collective decisions even when individual agents are imperfect or unreliable. The project seeks to develop novel methods for peer evaluation, collective reasoning, and trustworthy collaboration among AI agents. This research has significant potential to advance the foundations of multi-agent intelligence and contribute to reliable autonomous decision-making in complex environments. Dr. Du's expertise in language model reasoning and evaluation, together with his ongoing work on AI agents, positions him well to pursue this ambitious research agenda.

The University of Texas at Dallas is committed to providing sustained institutional support for Dr. Du's research and career development. The department has provided him with start-up funding, dedicated research space, and access to computing resources. We also support his research through a reduced teaching workload, support for graduate student recruitment, faculty mentoring, and professional development opportunities. These institutional investments will help Dr. Du further develop his independent research program and expand its impact. Our commitment extends beyond the proposed ONR award, reflecting our confidence in his long-term potential and our investment in his development as a leading faculty member.

Dr. Du exemplifies the qualities sought by the ONR Young Investigator Program: scientific creativity, technical excellence, research independence, and exceptional promise for sustained contributions to his

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field. I strongly endorse his application and reaffirm our department's commitment to supporting his continued success at The University of Texas at Dallas.

Sincerely,

Rym Zalila-Wenkstern



Rym Zalila-Wenkstern, Ph.D.
Professor and Interim Department Head
Department of Computer Science
The University of Texas at Dallas
Email: rymw@utdallas.edu
Phone: (972) 883 2091

8 Data Management Plan

Types of data, software, and other materials. The project will produce (1) three benchmarks of questions with known answers: math and code problems with injected shared mistakes (A), multi-document questions composed from event structures, including synthetic maritime scenarios (B), and scientific-claim and peer-review questions with inserted errors (C); (2) logs of agent answers, ratings, debates, tool calls and their outputs, judge rulings, and audit outcomes; (3) software: the agentic debate harness, the shared-factor reliability model, the impartial selection mechanism, the audit scheme, red-team agents, reinforcement-learning code, and the simulation code behind the theoretical results; and (4) fine-tuned weights of open-weight models, where the original licenses permit release.

How the data will be acquired. Benchmark A draws on public datasets (GSM8K and MATH, released under the MIT license) and public programming problems. Benchmark B is generated by our MEQA pipeline from event structures; its maritime scenarios are fully synthetic and unclassified. Benchmark C uses public papers and the public reviews and rebuttals posted on OpenReview, under the platform's terms, together with published reproducibility reports. Logs are produced by running agents built on open-weight models on UT Dallas computers and on commercial models through their interfaces. No human-subjects data are collected; in experiments, human audits are simulated by revealing known answers.

Time and location of acquisition. Not scientifically pertinent: all data are generated or collected computationally at UT Dallas during the 36-month project.

Processing. Questions are normalized to a common schema, deduplicated, and split into development, evaluation, and held-out sets by topic and by model family. Planted cases are generated by versioned scripts that record each edit and its effect on the answer. Logs are stored as produced and then summarized for analysis; every number in a publication is regenerated by a released script from the stored logs.

File formats and naming conventions. Questions and logs are stored as JSON Lines and Parquet; code is Python, released under the Mozilla Public License 2.0, as in the PI's earlier projects; model weights use the safetensors format. File names give the benchmark, split, and version (for example, `bmB_eval_v1.2.jsonl`), with semantic versioning and a change log for each release.

Quality assurance and quality control. The known answer of every generated question is confirmed by construction and by an independent script; two students review a random sample of each release, and disagreements are resolved before release. Each log record carries the model version, prompt version, random seed, and a hash of tool outputs, so that any debate can be re-run. The harness is frozen at each milestone, and automated runs confirm that frozen versions reproduce stored results.

Origin of existing data. GSM8K [17], MATH [32], the MEQA benchmark and pipeline [55], public OpenReview reviews, published reproducibility reports, and PeerEval logs produced by the PI's group. Each release records the source and license of every component.

Standards for data and metadata. Datasets will carry Croissant machine-learning metadata and a datasheet describing their purpose, composition, collection, and known limitations; code repositories will include environment files that pin software versions.

Preservation and access. Code will be public on GitHub, with each milestone release archived on Zenodo under a DOI. Benchmarks and logs will be released on Hugging Face and on project webpages linked from the PI's lab website, by the time of the associated publication, and kept available for at least five years after the project ends. Working copies stay on access-controlled lab machines, with secondary copies on departmental systems at least once per semester; HPC scratch storage is never used as an archive. Final reports will be submitted as required by DoD Instruction 3200.12.

Balance of value and cost. We will keep all data behind published results. Raw logs of exploratory runs, which can be large, will be kept for one year and then reduced to summaries, since the released code can regenerate them.

Security. All data are public or synthetic, unclassified, and contain no controlled unclassified information or personally identifiable information. If any data were later found to raise national security concerns, they would not be released, in accordance with DoD Instruction 5230.09.

References Cited

[1] A. K. Agogino and K. Tumer (2008). Analyzing and Visualizing Multiagent Rewards in Dynamic and Stochastic Domains. Autonomous Agents and Multi-Agent Systems 17(2):320–338.

[2] M. G. Allingham and A. Sandmo (1972). Income Tax Evasion: A Theoretical Analysis. Journal of Public Economics 1(3–4):323–338.

[3] E. S. Allman, C. Matias, and J. A. Rhodes (2009). Identifiability of Parameters in Latent Structure Models with Many Observed Variables. The Annals of Statistics 37(6A):3099–3132.

[4] N. Alon, F. Fischer, A. D. Procaccia, and M. Tennenholtz (2011). Sum of Us: Strategyproof Selection from the Selectors. In Proceedings of the 13th Conference on Theoretical Aspects of Rationality and Knowledge (TARK XIII), pp. 101–110.

[5] A. Amayuelas, X. Yang, A. Antoniades, W. Hua, L. Pan, and W. Y. Wang (2024). MultiAgent Collaboration Attack: Investigating Adversarial Attacks in Large Language Model Collaborations via Debate. Findings of EMNLP, pp. 6929–6948.

[6] S. Arnesen, D. Rein, and J. Michael (2024). Training Language Models to Win Debates with Self-Play Improves Judge Accuracy. arXiv:2409.16636.

[7] R. Avenhaus, B. von Stengel, and S. Zamir (2002). Inspection Games. In Handbook of Game Theory with Economic Applications, Vol. 3, Ch. 51, pp. 1947–1987. Elsevier.

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